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(54) **PORTABLE ACCESS POINT FOR SECURE
USER INFORMATION USING A
BLOCKCHAIN BACKED CREDENTIAL**

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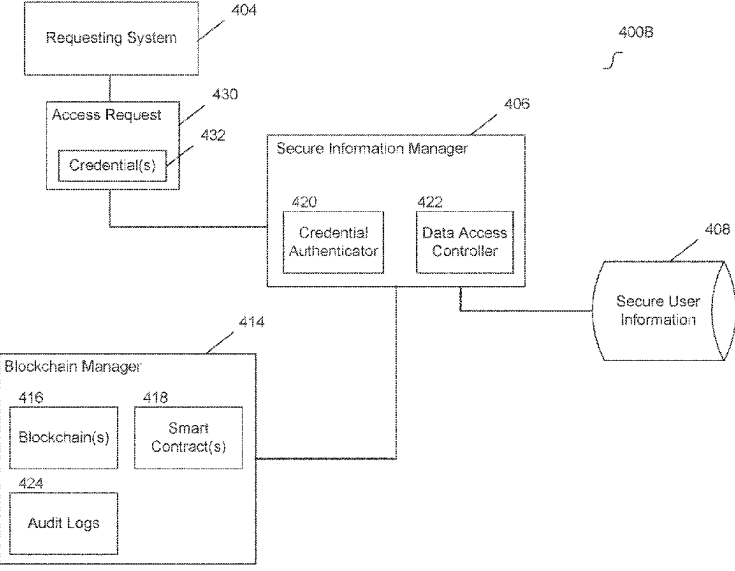
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(57) **ABSTRACT**

Embodiments permit scope limited access to a user's secure
information using blockchain backed credential(s). A user
can register with a secure information manager and control
the scope with which the user's secure information is shared.
For example, the user can permit a vetted entity access to the
user's secure information via a portable access point. The
user can select scope definition that control how the user's
secure information is shared with the vetted entity. The
vetted entity can scan the user's portable access point and
request a credential. The credential can be a blockchain
backed credential that is assigned access privileges that
correspond the user's selections. The vetted entity can then
issue data access request(s) using the credential. The secure
information manager can permit the vetted entity scope
limited access to the user's secure information that corre-
sponds to the access privileges assigned to the credential.

20 Claims, 10 Drawing Sheets



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(52) **U.S. Cl.**

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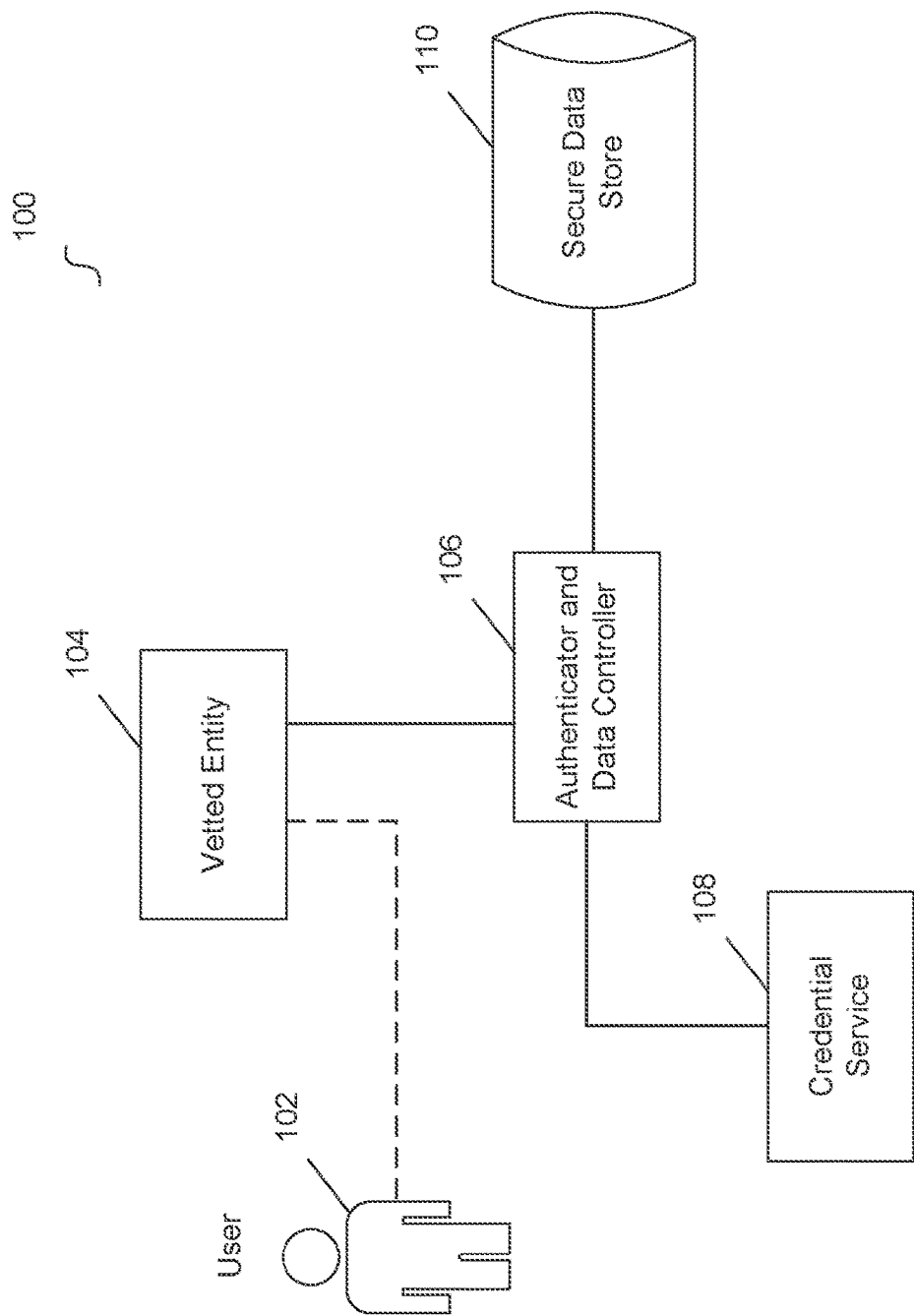


Fig. 1

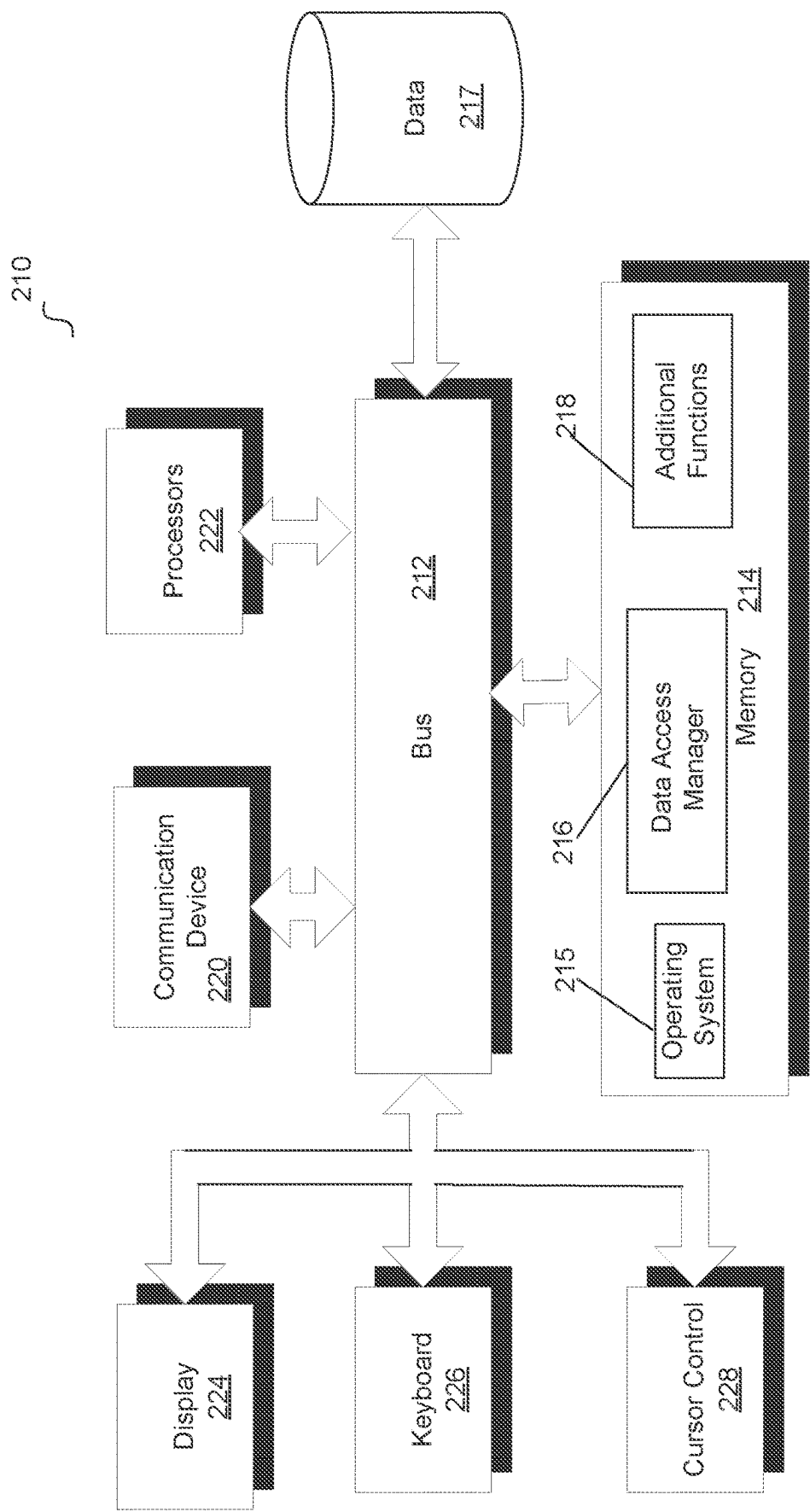


Fig. 2

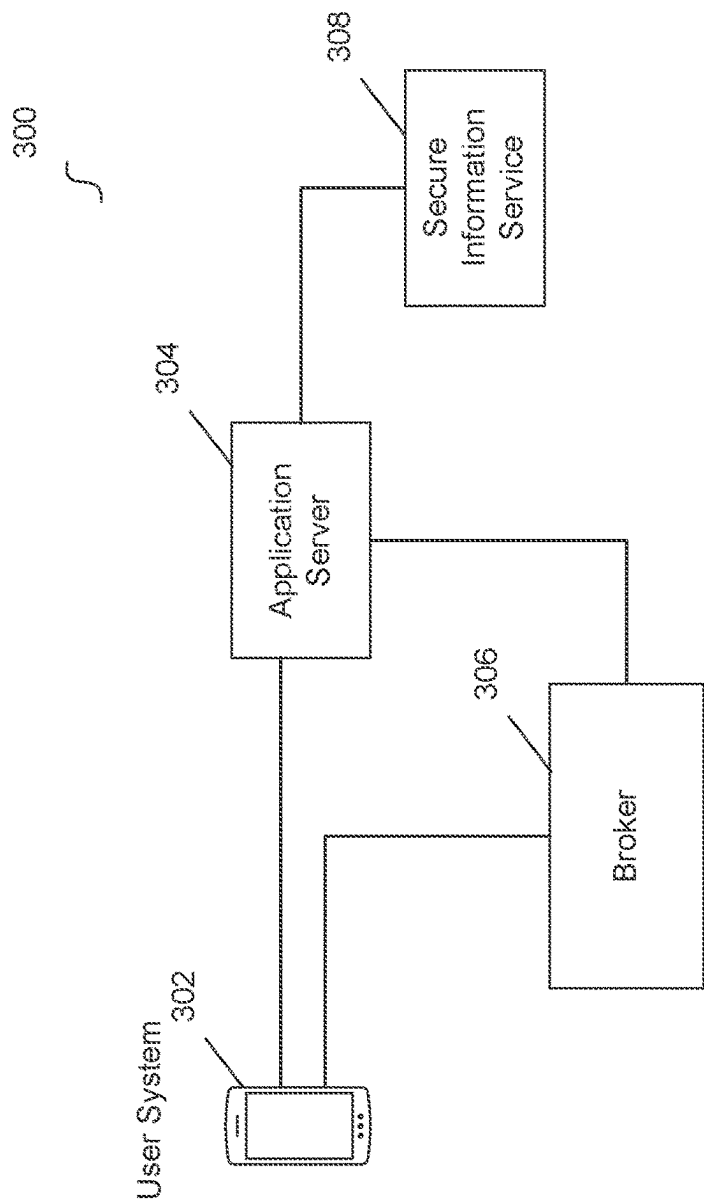


Fig. 3

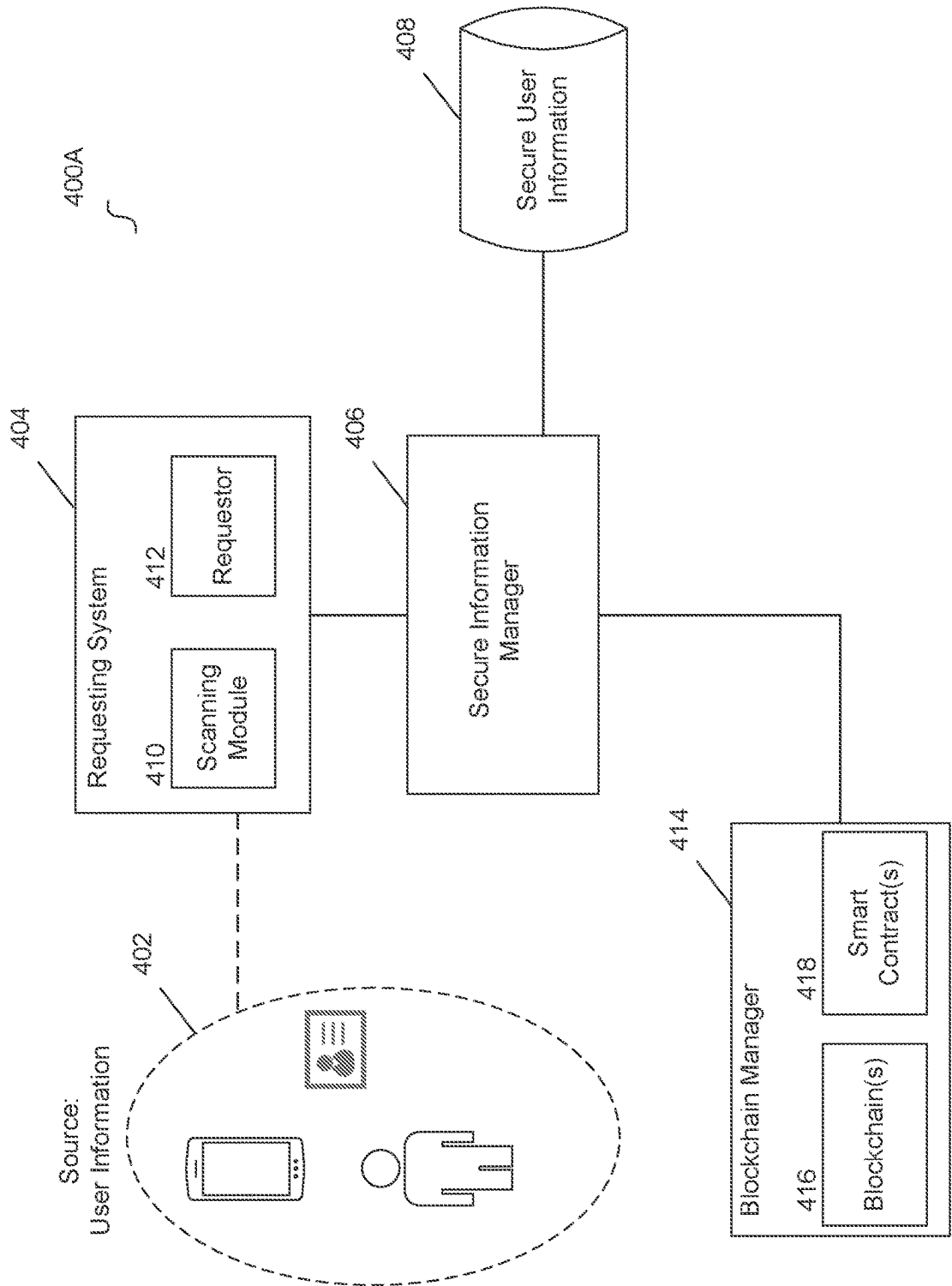


Fig. 4A

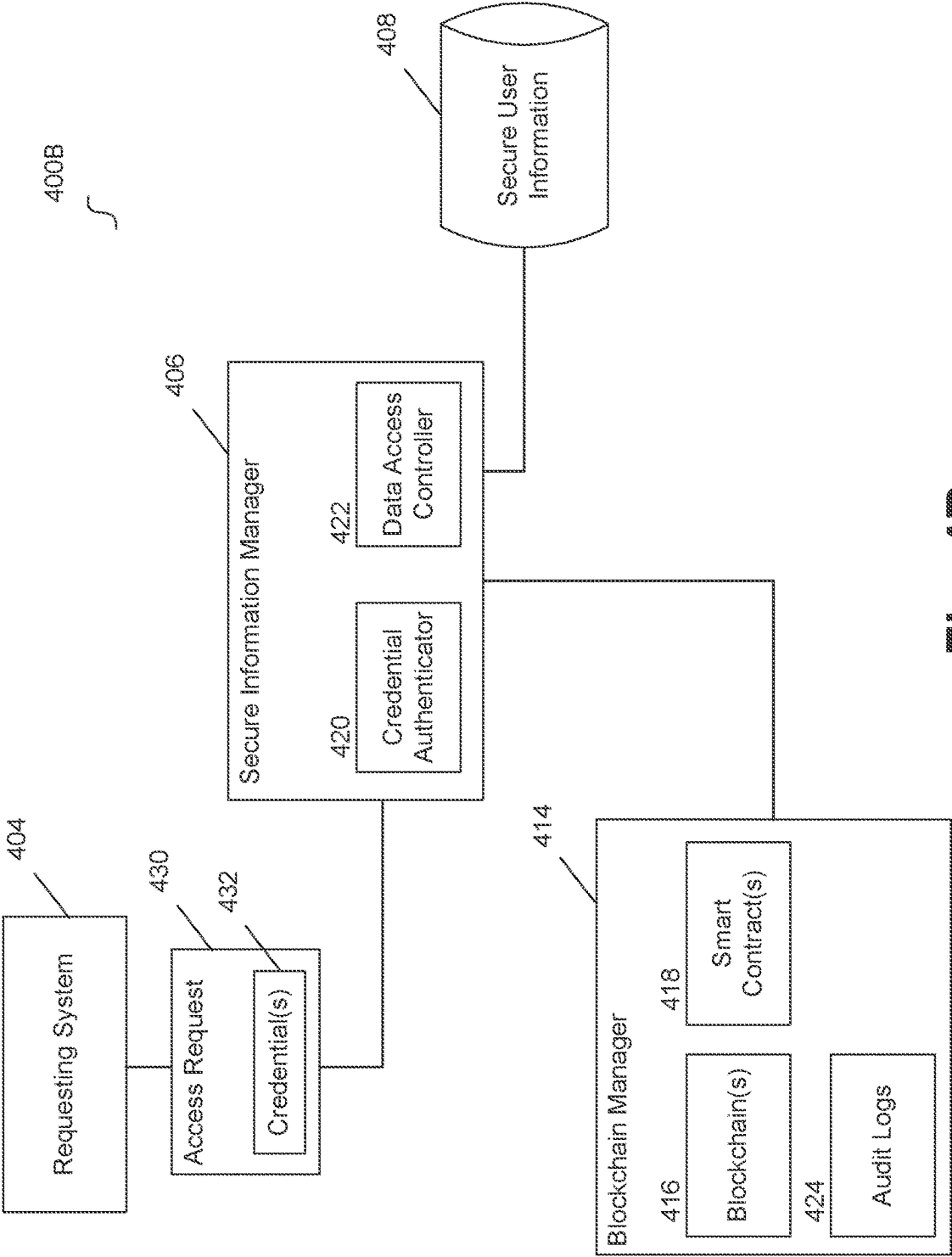


Fig. 4B

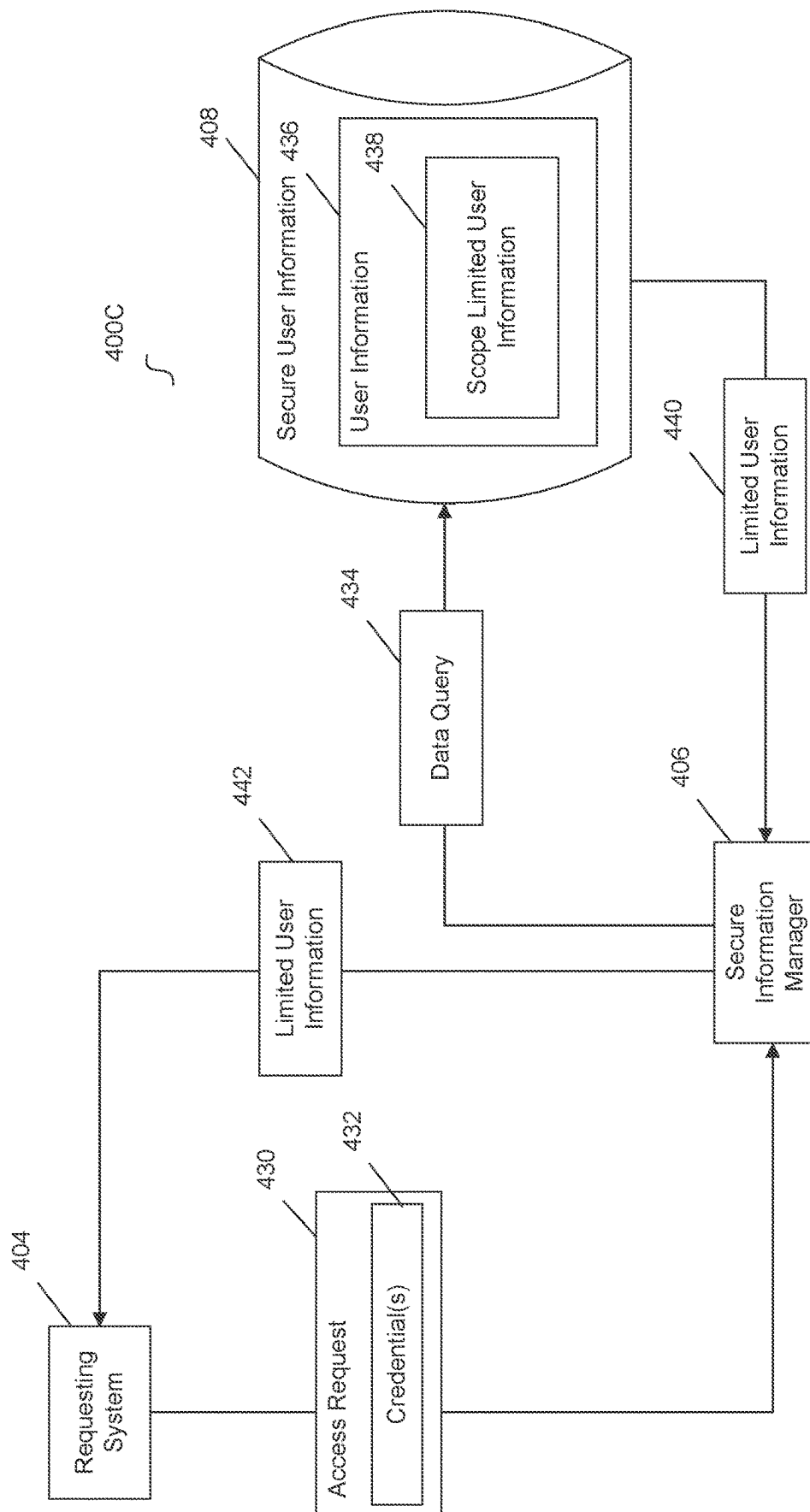


Fig. 4C

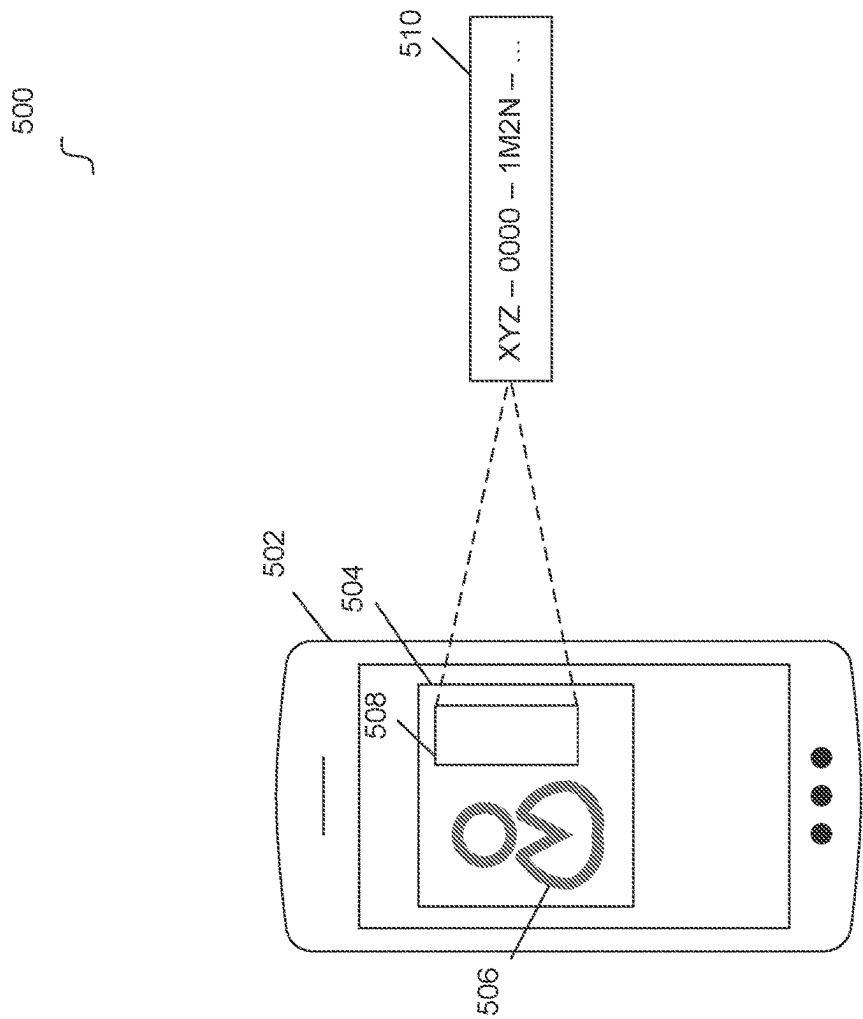


Fig. 5

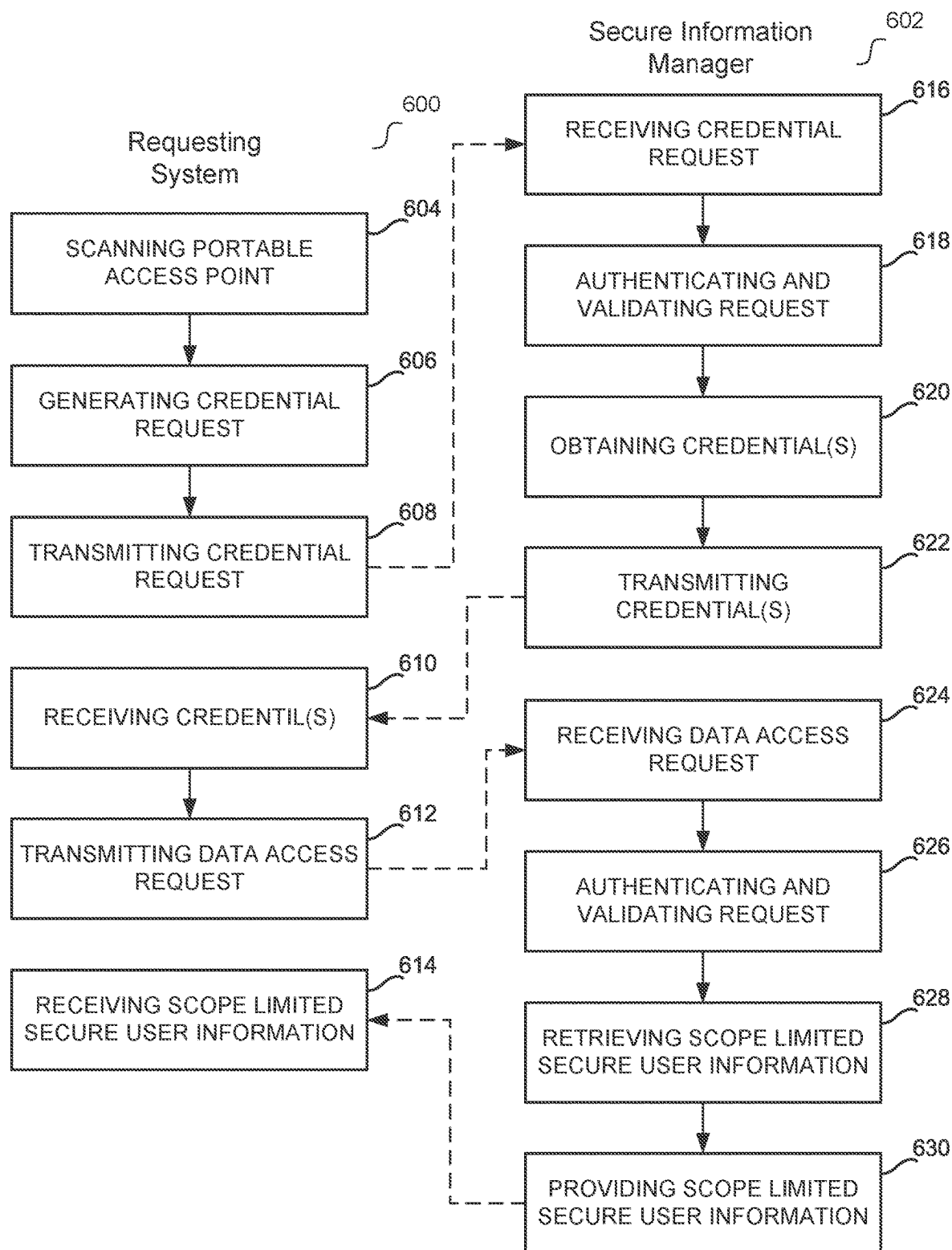
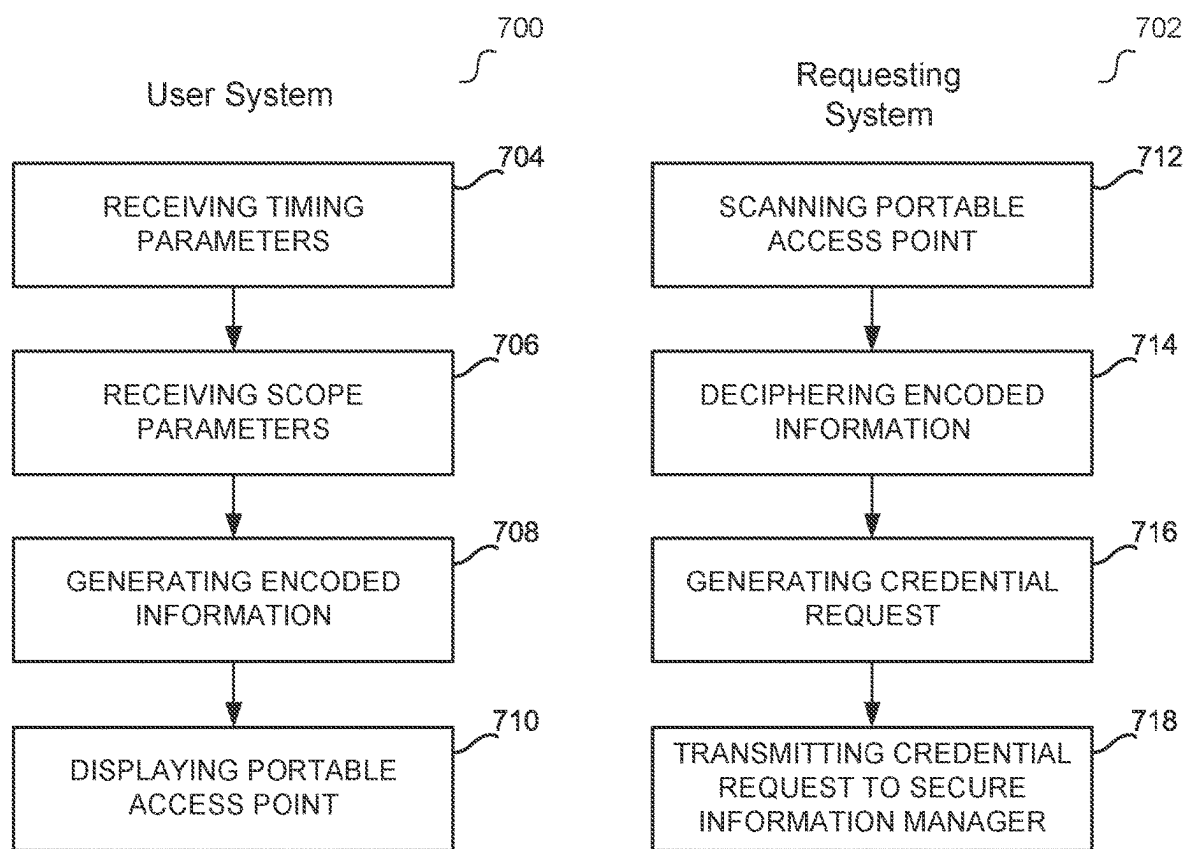
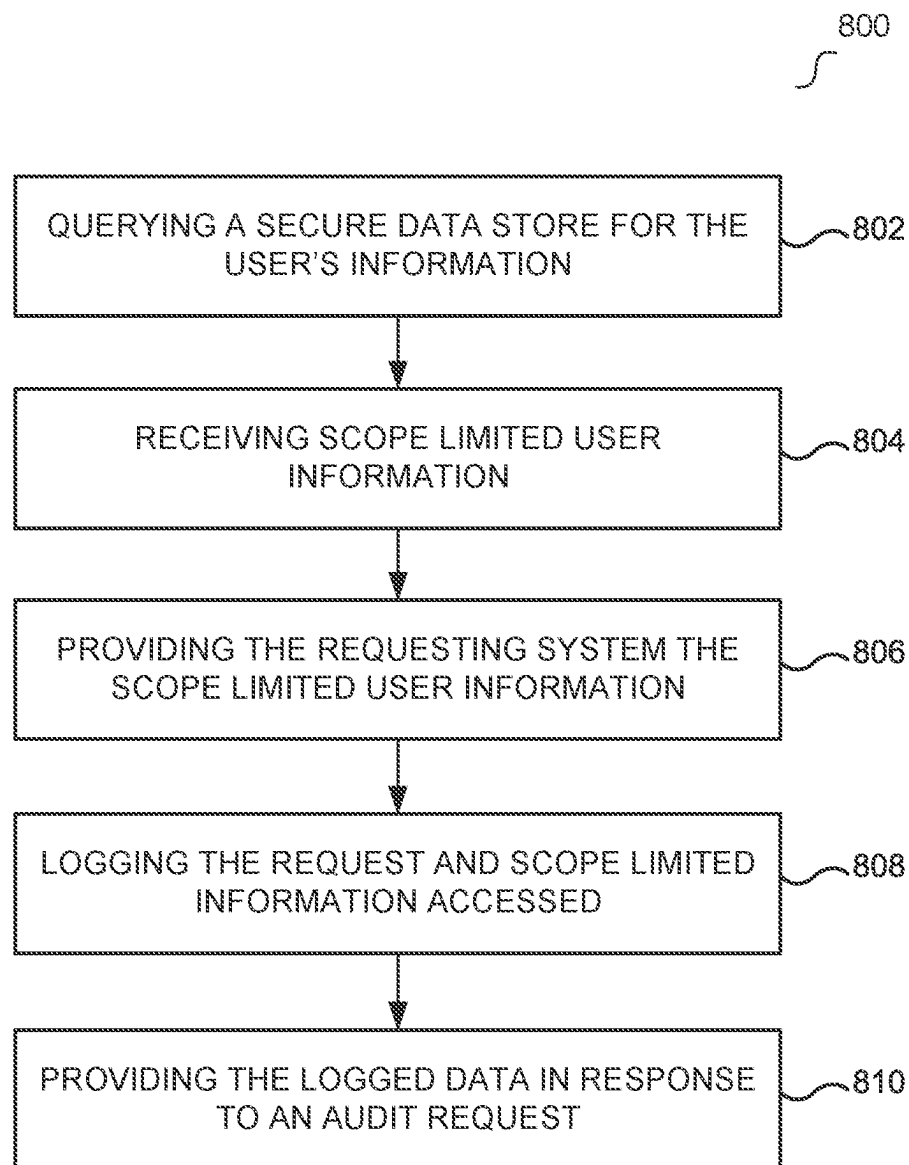


Fig. 6

**Fig. 7**

**Fig. 8**

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PORTABLE ACCESS POINT FOR SECURE USER INFORMATION USING A BLOCKCHAIN BACKED CREDENTIAL

FIELD

The embodiments of the present disclosure generally relate to secure storage system(s) that permit scope limited access to a user's secure information using blockchain backed credential(s).

BACKGROUND

The proliferation of computing and connected devices has generated vast amounts of data that requires management. As data grows in size, the technological challenges related to efficiently managing the data has become increasingly complex. For example, sharing secure data among multiple parties has been a longstanding problem in the field of data management. Security techniques that permit a user to manage secure information, such as authentication, validation, and permission workflows, can be cumbersome and, in some scenarios, impractical. Security protocols that achieve practical secure data sharing in scenarios that cause friction for traditional data sharing protocols can provide substantial value.

SUMMARY

The embodiments of the present disclosure are generally directed to systems and methods for permitting limited access to a user's secure information using credential authentication and user information verification. A credential request for one or more credentials that permit access to a user's secure information can be received at a secure information manager from a requesting system, the credential request comprising user identifying information, entity identifying information, and scope definitions, wherein the requesting system generates the credential request in response to scanning a portable access point of a user. The user identifying information and the entity identifying information can be validated at the secure information manager. A blockchain credential with access privileges that correspond to the scope definitions can be assigned to the entity and in response to the validating, wherein the blockchain credential assignment is recorded on a private blockchain that manages the blockchain credential. In response to one or more access requests from the entity that comprises the assigned blockchain credential, access to the user's secure information can be permitted that is limited to the access privileges of the blockchain credential.

Features and advantages of the embodiments are set forth in the description which follows, or will be apparent from the description, or may be learned by practice of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

Further embodiments, details, advantages, and modifications will become apparent from the following detailed description of the preferred embodiments, which is to be taken in conjunction with the accompanying drawings.

FIG. 1 illustrates a system for permitting scope limited access to a user's secure information using non-fungible tokens according to an example embodiment.

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FIG. 2 illustrates a block diagram of a computing device operatively coupled to a prediction system according to an example embodiment.

FIG. 3 illustrates a system registering users for secure information management according to an example embodiment.

FIGS. 4A, 4B, and 4C illustrate systems with a secure information manager that permit scope limited access to a user's secure information using blockchain credential(s) according to an example embodiment.

FIG. 5 illustrates a portable access point according to an example embodiment.

FIG. 6 illustrates a flow diagram for permitting limited access to a user's secure information using credential authentication and user verification according to an example embodiment.

FIG. 7 illustrates a flow diagram for defining access limitations for sharing a user's secure information according to an example embodiment.

FIG. 8 illustrates a flow diagram for retrieving scope limited user information from a secure data store and logging the access according to an example embodiment.

DETAILED DESCRIPTION

Embodiments permit scope limited access to a user's secure information using blockchain backed credential(s). A user can register with a secure information manager and control the scope with which the user's secure information is shared. For example, the user can permit a vetted entity (e.g., service provider, health care provider, other individual, etc.) access to the user's secure information via a portable access point. The user can select scope definition that control how the user's secure information is shared with the vetted entity. The vetted entity can scan the user's portable access point and request a credential that permits access to the user's secure information via the scanning. For example, the credential can be a blockchain backed credential that is assigned access privileges that correspond the user's selections.

The vetted entity can then issue one or more data access requests using the credential. For example, the data access request(s) can be authenticated and validated by the secure information manager. The secure information manager can permit the vetted entity scope limited access to the user's secure information (based on an authenticated and validated data access request) that corresponds to the access privileges assigned to the credential. The user can revoke the access privileges assigned to the credential and/or vetted entity at any time. The access privileges assigned to the credential can include an expiration timer, after which the credential will no longer authenticate with the secure information manager.

Embodiments achieve fine grained user controlled access to the user's secure information that is efficient and secure. For example, the user's portable access point is configured to efficiently define sharing conditions for the user's secure information. In addition, the issued credentials and secure information manager enforce the user's sharing conditions in a trusted manner. In embodiments where the credential is blockchain backed, the blockchain based management ensures that the credential is authentic and mitigates against fraudulent attempts to access the user's secure information.

Blockchain backed credentials can be issued to vetted entities. For example, the vetted entity can be an individual, organization, group of individuals, and the like. A vetted entity can undergo a vetting workflow, after which the entity

can be issued credential(s) to access a user's secure information. The vetting workflow can include one or more of: identity verification, credential verification (e.g., government credentials, medical credential, financial advisor credentials, etc.), cyber security validation, and any other suitable vetting. The vetted entity can generate a credential request for access to a user's secure information by scanning (via a computing system and scanning component) the user's portable access point.

A portable access point can be visual access point that, when scanned, can permit access to a user's secure information (e.g., stored and managed via a secure information manager). For example, the visual access point can comprise a visual representation of the user linked to the portable access point (e.g., facial image) and encoded information related to the scope of the user's secure information and/or permitted access.

The user can configure the portable access point and the encoded information displayed. For example, the portable access point can be displayed via an application executing at the user's wireless device (e.g., smartphone, tablet, etc.) and the user can interact with the application and select a sharing scope for the user's secure information. Embodiments of the portable access point are dynamic such that the user's selections via the application generate different versions of the portable access point with different encoded information displays. For example, the user can define a sharing scope that identifies data points of the user's secure information that can be shared with a vetted entity via scanning of the portable access point. The user can also define a sharing scope for a time period over which the user's secure information can be shared with a vetted entity via scanning of the portable access point.

One or more scanning element(s) of the vetted entity's computing system(s) can scan the portable access point and generate a credential request using the information from the portable access point. For example, the credential request can include: one or more entity credentials; identifying information for the user; scope definitions that define the access privileges for the requested credential relative to the user's secure information, and/or credential type. The entity credential(s) can include credentials issued to the entity (e.g., issued to one or more users and/or identities associated with the entity) after the entity is vetted by the vetting workflow. Example entity credential(s) include an access token (e.g., Security Assertion Markup Language (SAML), Open Authorization (OAuth), etc.), one or more cryptographic keys or signatures, and the like. The secure information manager can authenticate the entity credentials provided in a request prior to issuing access credential(s) to the vetted entity.

The credential request can also include user identifying information. Example user identifying information can be one or more of: a set of user data that identifies the user (e.g., full name, birthdate, home city, state, and/or zip code, physical appearance, etc.), an image of a government issued document that identifies the user (e.g., driver's license, passport, etc.), biometric information (e.g., fingerprints, eye scan, DNA information, etc.), and other suitable identifying information for the user.

Embodiments of the user's secure information can be electronic health data segmented based on parameters, and scope definitions that define the access privileges for the requested credential relative to the user's secure information can correspond to limited portions of the user's electronic health data. For example, the parameters can include: originating physician and/or medical organization (e.g., entity

identifier(s)), type of information (e.g., medications, tests and results, medical history, family history, biometrics, physician and patient communications, physician notes, vaccine information, allergies, etc.), relevant health practice (e.g., cardiology, primary care, neurology, oncology, etc.), date of information origination, electronic health record format, other Health Level Seven (HL7), Fast Healthcare Interoperability Resource (FHIR), and/or Substitute Medial Applications and Reusable Technologies (SMART) on FHIR data parameters, or any other suitable health data parameters. The user can define what portions of the user's electronic health data to share via the user's portable access point by providing parameter values that define the scope.

The secure information manager can validate and authenticate the credential request and obtain a credential from a blockchain service in response to the request. For example, the secure information manager can transmit the obtained credential to the vetted entity's computing system(s). The credential can be an NFT managed by a blockchain service, and the vetted entity's computing system(s) can comprise a token wallet affiliated with the vetted entity that stores the NFT. In other examples, the credential can be any other suitable blockchain based credential and can be stored in any suitable storage location by the computing system(s) of the vetted entity. Example blockchain backed credentials can include an access token (e.g., Security Assertion Markup Language (SAML), Open Authorization (OAuth), etc.) managed via a blockchain, one or more cryptographic keys managed via a blockchain, and the like.

After the credential is issued to the vetted entity, the vetted entity's computing system(s) can issue data access request(s) using the credential. For example, a data access request that includes the issued credential, an identifier for the vetted entity that issued the request, and user identifying information can be transmitted to the secure information manager. The secure information manager can validate and authenticate the data access request, retrieve scope limited secure user information in response to the data access request, and return the scope limited secure user information to the vetted entity's computing system(s).

The data access request can define one or more data points of the user's secure information. For example, the user's secure information can be electronic health data segmented based on parameters. The data access request can include specific parameter values that define the scope of the user's secure information requested. The secure information manager can retrieve secure user information that corresponds to the requested data point(s) included in the data access request. For example, the secure information manager can retrieve secure user information that correspond to a portion of the requested data point(s) included in the data access request. When the data access request includes user data point(s) outside the set of scope privileges of the vetted entity/provided credential(s), the secure information manager may retrieve only the portion of the user data point(s) covered by the set of scope privileges.

The vetted entity can comprise any suitable entity that performs services for the user, such as home services (e.g., home construction, repair, etc.), automobile services (e.g., repairing the user's car), medical services (e.g., medical services relates to a doctor's office, hospital, emergency room, first responders, etc.), financial services (e.g., accounting, trustee services, financial advising, etc.), technology services (e.g., system administrator services, web hosting, etc.), and the like. In an example, the user's secure information can be electronic health records and the vetted entity can be a health care provider requesting access to the

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user's electronic health records. In this example, the health care provider can scan the user's portable access point (e.g., via the user's wireless device) and request credential(s) from a secure information manager and a blockchain service. Once the credential request from the health care provider is authenticated and validated, the blockchain service can issue the health care provider a credential that provides the vetted entity scope limited access to the user's electronic health records.

The user can define which credential type to issue to the health care provider via the user's application executing at the user's wireless device. For example, the user can select one of a first credential (e.g., vanishing credential), second credential (episodic credential), and/or third credential (e.g., durable credential), and the application can display a version of the user's portable access point in response to the selection. The version of the user's portable access point can encode the credential type selected by the user for the health care provider. The health care provider can scan the portable access point and issue the credential request, which then includes the credential type the user selected for the health care provider.

The user can also define the access privileges for the credential requested via scanning of the portable access point using the application. For example, the user can select secure user information data points, segments of data points, parameter values used to group data points, or any other suitable definitions for portioning the user's secure information. The version of the user's portable access point can encode the access privileges for the credential defined by the user for the health care provider. The health care provider can scan the portable access point and issue the credential request, which then includes the access privileges the user defined for the health care provider.

The blockchain service and/or the secure information manager can then issue the health care provider a blockchain back credential that corresponds to the credential type the user selected and/or comprises the access privileges the user defined. Once the health care provider computing system receives the issued credential, the system can issue a data access request to the secure information manager to access the user's electronic health records. To permit the access, the secure information manager can authenticate the credential via smart contract call(s) to the blockchain service. When the credential authenticates, the secure information manager can permit the health care provider's system scope limited access to the user's electronic health records, such as access limited to the privileges assigned to the credential.

The health care provider's access using the credential(s) issued to the provider can be logged. For example, the blockchain service can log historic access to the user's electronic health records at one or more private blockchain(s). The health care provider's electronic health record access can be audited via these private blockchain(s) that store the logs.

Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the embodiments. Wherever possible, like reference numbers will be used for like elements.

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FIG. 1 illustrates a system for permitting scope limited access to a user's secure information using non-fungible tokens according to an example embodiment. Diagram 100 includes user 102, vetted entity 104, authenticator and data controller 106, credential service 108, and secure data store 120. Vetted entity 104 can issue a request to authenticator and validator 106 for access to user 102's secure information stored at secure data store 108. Vetted entity 104 can be any suitable person, group of people, organization or company, and the like that undergoes a vetting workflow.

Vetted entity 104 comprises computing system(s) associated with the vetted entity. For example, an application at the computing system(s) can permit a registered identity of vetted entity 104 to login to the application. Vetted entity 104 and one or more registered identities of the vetted entity can be registered with authenticator and data controller 106. For example, authenticator and data controller 106 can be part of a secure information manager that manages access to secure data store 110.

User 102 may be collocated (in the same physical location) as the computing system(s) of vetted entity 104. For example, the computing system(s) can obtain user information from user 102, such as via scanning a portable access point for user 102. A portable access point can be a visual access point for user 102's secure information. For example, the portable access point can include a depiction of user 102, such as a facial image, and encoded information, such as a QR code, barcode, a sequence of symbols (e.g., alphanumeric, hexadecimal, etc.), and any other suitable encoded information. The encoded information can represent: identifying information for user 102; scope definitions corresponding to user 102's secure information; time limitations for access to user 102's secure information; and other suitable information.

Vetted entity 104 can scan the portable access point for user 102 to generate a credential request for access to the user's secure information stored at secure data store 110. For example, user 102 and vetted entity 104 (e.g., computing systems for the vetted entity) can be collocated, the user's portable access point can be carried by user 102 (e.g., displayed via the user's wireless device). In other examples, user 102 and vetted entity 104 may be remote from one another.

Upon scanning user 102's portable access point, the computing system(s) of vetted entity 104 can issue a credential request to authenticator and data controller 106 for credential(s) that permit access to the user's secure information stored at secure data store 110. For example, the credential request can include user identifying information obtained from scanning the user's portable access point. Authenticator and data controller 106 can authenticate that the credential request originated via scanning of user 102's portable access point. For example, embedded information from the portable access point can be included in the request and authenticated by authenticator and data controller 106. Authenticator and data controller 106 can also authenticate that the requesting system corresponds to a vetted entity, such as vetted entity 104.

The credential request can also include scope definitions for the scope of the requested access (to the user's secure information). For example, the embedded information from the portable access point can include these scope definitions. Authenticator and data controller 106 can request credential(s) from credential service 108 after authentication of the credential request. The requested credential(s) can be assigned access privileges that correspond to the scope definitions provided in the credential request. Credential

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service **108** can issue the credential(s) to vetted entity **104**, such as credential(s) backed by a private blockchain managed at credential service **108**. The private blockchain can record the issuance of the credential(s) to vetted entity **104** (e.g., an identifier that represents vetted entity **104**, etc.). Authenticator and data controller **106** can receive the credential(s) from credential service **108** and provide the credential(s) to vetted entity **104**.

After receiving the credential(s), the system of vetted entity **104** can issue a data access request to authenticator and data controller **106** to access to user **102**'s secure information stored at secure data store **110**. For example, the data access request can include the issued credential(s), identifying information for user **102** (e.g., obtained via the portable access point, or any other suitable identifying information), and an identifier of vetted entity **104**.

Authenticator and data controller **106** can authenticate the credential(s) included in the data access request and validate the user's identifying information. For example, authenticator and data controller **106** can authenticate that the data access request credential(s) correspond to one or more credentials issued to vetted entity **104**. Authenticator and data controller **106** can validate the credential(s) via credential service **108**. For example, credential service **108** can include a private blockchain service that manages credential(s) related to user **102** and permissions to access user **102**'s secure information stored at secure data store **110**. Authenticator and data controller **106** can issue one or more application programming interface (API) calls (e.g., smart contract calls) to credential service **108**.

In response to these API call(s), credential service **108** can authenticate that the provided credential(s) are: assigned to vetted entity **104**; and correspond to defined scope permissions for user **102**'s secure information. A private blockchain managed by credential service **108** can include an immutable ledger that records information for assigned credential(s). For example, the private blockchain can record identifying information for the user whose secure information is scoped by a given credential (e.g., user **102**), the scope definitions that correspond to the given credential, an entity assigned the given credential, entity assignment changes for the given credential, and the like.

Credential service **108** can authenticate the provided credential(s) against the private blockchain to confirm that vetted entity **104** is assigned the credential(s). Credential service **108** can also provide authenticator and data controller **106** scope definitions recorded at the private blockchain that correspond to the credential(s) provided in the data access request. For example, the scope definitions can define the portion(s) of the user's secure information that the provided credential(s) and vetted entity **104** are authorized to access.

Authenticator and data controller **106** can also validate that the user's identifying information from the data access request corresponds to a registered person that has secure information stored at secure data store **108**. For example, users can register with authenticator and data controller **106**, and the secure information of registered users can be stored at secure data store **110**. An application service can provide portable access point(s) for registered users so that vetted entity **104** can scan the portable access point(s) to request access to the registered users' secure information stored at secure data store **110**.

In response to the authentication of the provided credential(s) and vetted entity **104** and the validation of user **102**'s identifying information, authenticator and data controller **106** can permit vetted entity **104** scope and time limited

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access to the user's secure information stored at secure data store **110**. For example, the scope and time limitations can be controlled by the scope permissions granted to the provided credential(s). The scope of the user's secure information can be limited to a relationship between vetted entity **104** and user **102**, or other suitable characteristics of vetted entity **104**. In another example, the access can be limited to a duration of time (e.g., days, weeks, months, etc.), after which authenticator and data controller **106** will no longer permit vetted entity **104** access unless another request is issued that includes credential(s) that authenticate via credential service **108**.

FIG. 2 is a block diagram of a computer server/system **210** in accordance with embodiments. As shown in FIG. 2, system **210** may include a bus device **212** and/or other communication mechanism(s) configured to communicate information between the various components of system **210**, such as processor **222** and memory **214**. In addition, communication device **220** may enable connectivity between processor **222** and other devices by encoding data to be sent from processor **222** to another device over a network (not shown) and decoding data received from another system over the network for processor **222**.

For example, communication device **220** may include a network interface card that is configured to provide wireless network communications. A variety of wireless communication techniques may be used including infrared, radio, Bluetooth®, Wi-Fi, and/or cellular communications. Alternatively, communication device **220** may be configured to provide wired network connection(s), such as an Ethernet connection.

Processor **222** may include one or more general or specific purpose processors to perform computation and control functions of system **210**. Processor **222** may include a single integrated circuit, such as a micro-processing device, or may include multiple integrated circuit devices and/or circuit boards working in cooperation to accomplish the functions of processor **222**. In addition, processor **222** may execute computer programs, such as operating system **215**, migration prediction component **216**, and other applications **218**, stored within memory **214**.

System **210** may include memory **214** for storing information and instructions for execution by processor **222**. Memory **214** may contain various components for retrieving, presenting, modifying, and storing data. For example, memory **214** may store software modules that provide functionality when executed by processor **222**. The modules may include an operating system **215** that provides operating system functionality for system **210**. The modules can include an operating system **215**, data access manager **216**, as well as other applications modules **218**. Operating system **215** provides operating system functionality for system **210**. Data access manager **216** may provide system functionality for permitting scope limited access to a user's secure information to a vetted entity, or may further provide any other functionality of this disclosure. In some instances, data access manager **216** may be implemented as an in-memory configuration.

Non-transitory memory **214** may include a variety of computer-readable medium that may be accessed by processor **222**. For example, memory **214** may include any combination of random access memory ("RAM"), dynamic RAM ("DRAM"), static RAM ("SRAM"), read only memory ("ROM"), flash memory, cache memory, and/or any other type of non-transitory computer-readable medium.

Processor **222** is further coupled via bus **212** to a display **224**, such as a Liquid Crystal Display ("LCD"). A keyboard

226 and a cursor control device 228, such as a computer mouse, are further coupled to communication device 212 to enable a user to interface with system 210.

In some embodiments, system 210 can be part of a larger system. Therefore, system 210 can include one or more additional functional modules 218 to include the additional functionality. Other applications modules 218 may include the various modules of Oracle® Health, Oracle® Data Integrator, Oracle® Cloud Infrastructure, Oracle® Autonomous Database, Oracle® Cerner®, Oracle® Cerner® Millennium, Oracle® Cerner® HealtheIntent, Oracle® Cerner® Seamless Exchange, Oracle® Cerner® Health-eCare, Oracle® Blockchain and Oracle® Cerner® HealtheLife and representative products across the Oracle® Health & Artificial Intelligence platform for example. A database 217 is coupled to bus 212 to provide centralized storage for modules 216 and 218 and to store, for example, registered person validation information, vetted entity information, authentication and validation related information, etc. Database 217 can store data in an integrated collection of logically-related records or files. Database 217 can be an operational database, an analytical database, a data warehouse, a distributed database, an end-user database, an external database, a navigational database, an in-memory database, a document-oriented database, a real-time database, a relational database, an object-oriented database, Hadoop Distributed File System (“HDFS”), disaster recovery database, backup database, or any other database known in the art.

Although shown as a single system, the functionality of system 210 may be implemented as a distributed system. For example, memory 214 and processor 222 may be distributed across multiple different computers that collectively represent system 210. In one embodiment, system 210 may be part of a device (e.g., smartphone, tablet, computer, etc.).

In an embodiment, system 210 may be separate from the device, and may remotely provide the described functionality for the device. Further, one or more components of system 210 may not be included. For example, for functionality as a user or consumer device, system 210 may be a smartphone or other wireless device that includes a processor, memory, and a display, does not include one or more of the other components shown in FIG. 2, and includes additional components not shown in FIG. 2.

A user can complete a registration workflow for secure information management. For example, the user can register with a secure information manager. Registered users can share their secure information via portable access point(s) that permit a vetted entity scope limited access to the registered user's secure information. FIG. 3 illustrates a system for registering users for secure information management according to an example embodiment.

Diagram 300 includes user system 302, application server 304, broker 306, and secure information service 308. User system 302 can be any suitable user client device, such as a smartphone, laptop, tablet, and the like. Application server 304 can be any suitable computing device that hosts an application, such as an application displayed to a user via user system 302. The application can be a web application, native application, any combination of these, or any other suitable application.

The user, via user system 302, can interact with the application to register an account. By generating a registered account, the user permits storage and management of the user's secure information by secure information service 308. For example, the registration can configure a user account

linked to the user and the user's secure information stored and managed by secure information service 308.

User system 302 can interact with application server 304 and/or broker 306 to complete the registration workflow. For example, application server 304 and/or broker 306 can be separate from the secure information service 308 in some embodiments. The registration of user(s) via a separate third-party entity (e.g., application server 304 and/or broker 306) can add a level of integrity and trust to the registration workflow.

The user's secure information can comprise electronic health records. In this example, the third-party entity can be a government entity, non-profit entity, coalition entity comprised of several individual entities, or any other suitable entity that supports a transparent and trustworthy registration workflow that links electronic health records to users and supports the secure storage of such electronic health records by secure information service 308. In these examples, the third-party entity can act as an intermediary that validates the users' identities, the defined scope(s) of the users' electronic health records stored by secure information service 308, and any other suitable aspects of the users' identity and secure information. The third-party entity can also support connections among different secure information services to ensure the users' electronic health records are available for user inspection and available to the users' medical providers.

Application server 304 can generate a unique link or registration code for each user (e.g., delivered to user system 302). Using the unique link or code, a user can access the application hosted by application server 304 via user system 302 and perform a registration workflow. User system 302 can receive a public key to register the user's account. The registration workflow can include the user generating a private key. For example, the private key can be used to manage the user's account and secure information. Any other suitable credentials (e.g., username and password, two factor authentication addresses, shared secret, etc.) can be issued to the user.

The registration workflow can include providing user identifying information such as first name, middle name, last name, date of birth, home address, phone number, a photo (e.g., facial image), employment history, city where the user was born, social security number, a user identifier for one or more electronic health records (e.g., identifier for a master patient index), gender, gender identity, ethnicity, and other suitable user information. The user can also provide their biometric information (e.g., thumbprint or eye scan) via a biometric scanner to improve the security measures and trustworthiness of the registration workflow and management of the user's secure information.

Broker 306 can validate the user's identifying information and link the registered account created for the user with the user's validated identity. For example, broker 306 can validate that the registered account is permitted to store electronic health records of a person that is validated as the user identity. This validation can prevent fraudulent attempts to access another person's secure information.

Once application server 304 and/or broker 306 perform a first portion of the registration workflow (e.g., receive user identifying information and validate the user's identity), secure information service 308 can generate account components for the user. The account components can include one or more portable access points for the user's secure information, relationship definitions for the user (e.g., defined relationship with other registered users or individuals), and any other suitable account components.

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Secure information service **308** can aggregate the user's secure information as part of (or after) the registration workflow. For example, secure information service **308** can retrieve, obtain, or otherwise aggregate the secure information that user authorized secure information service **308** to manage, such as the user's electronic health records.

The user's registered account can correspond to one or more portable access points. For example, the portable access point(s) can include a visual representation of the user (e.g., image) and encoded information about the user's secure information, such as scope definitions for sharing the user's secure information. FIG. 5 illustrates portable access point(s) according to embodiments.

Once the registration workflow is complete, the user can login to the generated account via user system **302** and application server **304**. The application can permit the user to define a scope for sharing the user's secure information via a generated portable access point. For example, the user can select one or more data points of the user's secure information, one or more default secure information data profiles, or define a scope for sharing the user's secure information in any other suitable manner.

The application can generate a portable access point that corresponds to the defined scope and display the portable access point at user system **302**. A computing system for an external entity, such as a vetted entity, can scan the portable access point to configure access to the user's secure information according to the defined scope. FIGS. 4A, 4B, and 4C illustrate systems with a secure information manager that permit scope limited access to a user's secure information using non-fungible tokens according to an example embodiment.

Diagram **400A** includes source **402**, requesting system **404**, secure information manager **406**, secure user information **408**, scanning module **410**, requestor **412**, blockchain manager **414**, blockchain(s) **416**, and smart contract(s) **418**. Requesting system **404** can be a system of a vetted entity, such as an entity that performed a vetting workflow by secure information manager **406**. Source **402** can include sources of user information, such as the user's body, user identification documents (e.g., driver's license, passport, etc.), the user's wireless device (e.g., smartphone), a portable access point for the user, and other suitable source(s) of user information.

Requesting system **404** can obtain user information from source **402**. For example, scanning module **410** of requesting system **404** can scan the user's portable access point. The user's portable access point can be displayed via the user's wireless device. For example, an application (e.g., native application, web application, etc.) executing at the user's wireless device can permit the user to configure a portable access point for sharing the user's secure information. FIG. 5 illustrates portable access point(s) and defining scope restrictions for the portable access point(s) via the application according to embodiments. The user's portable access point can be displayed by the user's wireless device on the lock screen (e.g., prior to inputting a password), in the wireless device's background, as a stored image in the wireless device's photo library, or via any other suitable display means.

Scanning module **410** can be a specialized device configured to scan the portable access point and obtain the identifying user information for a request, such as a wireless device that comprises an image capturing component (e.g., camera) and that executes an application configured to decipher the portable access point. Scanning module **410**

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can obtain user identifying information from any other suitable elements of source **402**.

Source **402** can include a wireless device of a person that comprises a predefined relationship with the user. For example, the user can be a registered person for whom secure information manager **406** manages secure user information **408**. The user can also register predefined relationship(s) with secure information manager **406**, such as personal relationships (e.g., parents, spouse, children, friends, etc.), professional relationships (e.g., co-manager of an enterprise application, etc.), or other suitable relationships.

The person(s) that comprise predefined relationships with the user can possess wireless device(s) that can provide requesting system **404** identifying information for the user. For example, any suitable user identifying information provided by the user's wireless device (e.g., the user's portable access point) can be provided to scanning module **410** via the wireless device of a person comprising a predefined relationship with the user. The related persons' wireless device(s) comprise an application that is registered with secure information manager **406** and that provides the user identifying information.

Requesting system **404** and requestor **412**, after scanning source **402**, can issue a credential request to secure information manager **406** for one or more access credential(s) that permit access to the user's secure information. For example, the requested credential(s) can be scope limited to the user identified by the user identifying information (e.g., obtained via sources **402**). The requested credential(s) can permit requesting system **404** access to secure information **408** via secure information manager **406**.

The credential request can also include user identifying information obtained via scanning module **412** (e.g., via scanning the user's portable access point), and any other suitable information. User identifying information can include full name, birthdate, social security number, local address and/or zip code, medical information (e.g., primary care physician, etc.), biometric information (e.g., fingerprints, eye scan, etc.), identifier for the user's medical health records (e.g., master-patient index identifier, medical record number, insurance claim identifier, etc.), vehicle make, model, and/or license plate, representations of government issued identification (e.g., driver's license photo), images of the user's face, and the like.

The user identifying information can be an identifier (e.g., alphanumeric sequence) deciphered from the user's portable access point (e.g., by scanning module **410**) and any other suitable user identifying information obtained by the scanning of the user's portable access point. The credential request can include a credential type. For example, the user can select (via the application executing at the user's wireless device) a credential type (e.g., credential type that comprises a corresponding expiration timer) and the application can generate a portable access point that encodes the credential type. Scanning module **410** can then decipher the credential type and requester **412** can include the credential type in the credential request.

In some embodiments, in order for requesting system **404** and requestor **412** to generate the credential request, the user may be prompted by requesting system **404** to enter a username and password, passcode, key, PIN, or any other suitable secure user code. For example, during a registration workflow the user may establish a username and password, passcode, key, PIN, etc. An application at requesting system **404** may be linked to secure information manager **406**, an honest broker, or any other suitable entity that can validate the user's username and password, passcode, key, PIN, etc.

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Once the application at requesting system **404** confirms that the secure user code is valid, the application can permit requesting system **404** and requestor **412** to generate the credential request.

To grant the credential request from requesting system **404**, secure information manager **406** can authenticate that requesting system **404** comprises a vetted entity. For example, the credential request from requester **412** can include one or more entity credentials issued to the vetted entity associated with requesting system **404**. The entity credential(s) can include an access token (e.g., Security Assertion Markup Language (SAML), Open Authorization (OAuth), etc.), one or more cryptographic keys or signatures, and the like. Secure information manager **406** can authenticate the entity credential(s) included in the credential request and validate user identifying information included in the credential request. For example, secure information manager **406** can authenticate that the entity credential(s) from the request authenticate as credential(s) issued to the vetted entity.

Secure information manager **406** can validate the user identifying information provided in the credential request. For example, secure information manager **406** can validate that the credential request was generated by scanning the user's portable access point. The validation can include: matching the identifier deciphered from the portable access point with a stored identifier for a registered user; validating that the additional user identifying information provided in the request validates against the user identifying information stored for the matched registered user; validating a credential type included in the credential request; and/or any other suitable validation.

When the entity credential(s) are authenticated and the user identifying information is validated, secure information manager **406** can issue a software call for an access credential on behalf of requesting system **404** (in association with the vetted entity) to blockchain manager **414**. For example, blockchain manager **414** can manage credentials that comprise access privileges for the user's secure information.

Blockchain manager **414** can manage blockchain(s) **416** for secure information manager **406**. For example, access credential(s) provided to vetted entities (in response to authenticated and validated credential requests) can include credentials managed by blockchain manager **414**. The credential(s) can be linked to a user's secure information and can be assigned varying levels of scope permissions for accessing the user's secure information. Blockchain(s) **416** can comprise blockchain ledger(s) that record credential transactions.

A blockchain is a list of records, each called a block, which can be linked through cryptography. In some blockchain implementations, each block includes a timestamp, a hash of the previous block, and transaction data. The timestamp proves that the transaction data was included when the block was added in order to get its hash. Because each block specifies the block previous to it, the set of blocks make a chain, with each new block reinforcing the set of blocks before it in the chain. Therefore, blockchains can be difficult to modify because data, once added to the blockchain, cannot be altered without altering subsequent blocks.

Credential(s) managed by blockchain manager **414** can comprise blockchain-backed identifiers specifying a unique or non-unique item that comprises access privileges for a user's secure information. Through a distributed ledger (e.g., blockchain), the assignment or ownership of the credentials can be tracked and verified. Blockchain manager **418** can manage blockchain(s) **416** and include smart contract(s) **418**

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that execute in combination with the blockchain(s) **416**. Blockchain(s) **416** can be Hyperledger blockchain(s), or any other suitable blockchain(s).

Blockchain manager **414** can manage different credentials that permit varying levels of access privileges to the user's secure information. For example, a data access request from a vetted entity can be authenticated via blockchain manager **414** and smart contract(s) **418** against blockchain(s) **416** to ensure that: the credential is managed by blockchain manager **414**, the credential was issued to the vetted entity that sent the data access request, and/or the credential is not expired. Diagram **400B** of FIG. **4B** includes an example data access request **430** that includes a credential issued by blockchain manager **414**.

Blockchain manager **414** can include a generating component that generates the credential(s) that permit varying levels of access privileges to the user's secure information. Each generated credential comprises a unique or non-unique long string identifier that identifies the credential. A generated credential can comprise information that links the credential to the user whose secure information that the credential is assigned privileges to access. The generated credential can include any suitable user identifying information described herein. Some example credential include blockchain based tokens, such as non-fungible tokens, semi-fungible tokens, tokens that can comprise several copies, non-unique tokens, or any other suitable credentials.

Blockchain manager **414** can issue one or more credentials to the vetted entity depending on the smart contract call from secure information manager **406**. For example, a first credential (e.g., vanishing credential) can provide access to a user's secure information for a limited duration of time (e.g., hours, days, weeks, etc.). In this example, once the limited duration of time expires, the first credential will no longer authenticate via blockchain manager **414** and blockchain(s) **416**. The user can define the expiration timer (e.g., via the application loaded at the user's wireless device) and the definition can be included in the user's portable access point. This defined timer can be included in the credential request issued by requesting system **404** via scanning of the user's portable access point. Secure information manager **406** can then include the defined timer in a smart contract call to blockchain manager **414**.

In another example, a second credential (e.g., episodic credential) can provide access to a user's secure information for a predefined duration of time (e.g., hours, days, weeks, etc.). In this example, once the predefined duration of time expires, the second credential will no longer authenticate via blockchain manager **414** and blockchain(s) **416**. In another example, a third credential (e.g., durable credential) can provide access to a user's secure information permanently or until the credential is explicitly revoked.

For example, the user's secure information can be electronic health records and the vetted entity can be a health care provider requesting access to the user's electronic health records. In this example, the health care provider can scan the user's portable access point (e.g., via the user's wireless device) and request credential(s) from blockchain manager **414**. Once the credential request from the health care provider is authenticated and validated, the blockchain manager can issue the health care provider a credential.

The user can define which credential to issue to the health care provider via the user's application executing at the user's wireless device. For example, the user can select one of the first credential, second credential, and/or third credential, and the application can display a version of the user's portable access point in response to the selection. The

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version of the user's portable access point can encode the credential selected by the user for the health care provider. The health care provider can scan the portable access point and issues the credential request, which then includes the credential the user selected for the health care provider.

In this example, the user may select the first credential (and define a timer) when the user expects that the healthcare provider will need limited access to the user's electronic health records for a short period of time, such as an urgent care facility visit. In another example, the user may select the second credential when the user expects that the healthcare provider will need access to the user's electronic health records for a moderate period of time, such as a surgery at a hospital the user does not frequent, a specialist visit that the user expects to not reoccur, and the like. In another example, the user may select the third credential when the user expects that the healthcare provider will need access to the user's electronic health records for a long or indefinite period of time, such as when the health care provider is the user's primary care physician, designated specialist for a long term health issue, and the like.

While the above examples describe a vetted entity that is a health care provider, the vetted entity can be another individual, such as an individual with a predefined relationship with the user (e.g., family member, close friend, etc.). In this example, the user may select the third credential to ensure that the individual maintains access to the user's electronic health records. After secure information manager **406** issues the one or more smart contract calls, blockchain manager **414** can issue one or more credentials to the vetted entity based on the smart contract call(s).

Blockchain manager **414** can generate credential(s) on-demand and/or store a plurality of pre-generated credential(s) affiliated with a user. For example, a credential can be generated on-demand in response to a credential request that comprises: user identifying information, scope privileges that correspond to the credential request, timing restrictions and/or a credential version (e.g., vanishing, episodic, and/or durable), or any other suitable information from the credential request. The generated credential can then be assigned to the vetted entity identifier from the credential request, and the assignment can be recorded to blockchain(s) **416**. The on-demand generated credential can be crafted for the specific parameters of the credential request. For example, the credential request can be generated in response to user selections at an information management application and/or scanning of a user's portable access point.

In another example, blockchain manager **414** can assign a pre-generated credential to the entity identified in the credential request. For example, pre-generated credential(s) can: include predefined scope privileges; correspond to a specific credential version (e.g., vanishing, episodic, and/or durable); comprise user identifying information that links the credential to the user; and/or include any other suitable information. Blockchain manager **414** can pre-generate, for a given user, multiple credentials with different combinations of parameters. For example, multiple credentials of each credential version (e.g., vanishing, episodic, and/or durable) can be pre-generated, where the multiple credentials of a given credential version can be assigned different predefined scope privileges. Because pre-generated credentials are not tailored to the specifics of a given credential request, the many different versions of the pre-generated credentials can accommodate many different credential requests.

For example, the different predefined scope privileges can correspond to: different scope templates the user has defined

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relative to the user's secure information (e.g., predefined groupings or segments of the user's electronic health data); different templates based on feedback from the vetted entities on what portion(s) of the user's secure information the entities require; a global scope that permits widescale access to the user's secure information; or any other suitable predefined scope privileges. Secure information manager **406** and/or blockchain manager **414** can match one of these pre-generated credentials to the credential request and assign the matching pre-generated credential to the vetted entity identified in the credential request.

The pre-generated credentials can also comprise a blank set of scope privileges, and secure information manager **406** can assign scope privileges to the credential that matches the credential request on-demand. Thus, a pre-generated credential can be matched to the credential request using credential version, and the matching credential can be assigned scope privileges specifically tailored to the credential request. Secure information manager **406** can communicate the privilege assignments to blockchain manager **414**, which can append the privilege assignments to blockchain(s) **416**.

Some credential requests from requesting system **404** can be generated by scanning a user's portable access point, such as a portable access point controlled by an information management application at the user's wireless device. The information management application can generate portable access points that correspond to pre-generated credentials. For example, the information management application can present the user a limited set of scope restrictions and/or a limited set of timing constraints (e.g., credential versions), where each combination of scope restriction and timing constraint corresponds to one of the pre-generated credentials at blockchain manager **414**. In this example, secure information manager **406** and/or blockchain manager **414** can match each credential request to the pre-generated credentials that corresponds to the credential request.

Upon receiving user selections and displaying a portable access point for a user, the information management application at the user's wireless device can also provide secure information manager **406** an indicator that identifies the particular portable access point (e.g., particular combination of scope restrictions and timing constraints) and a time at which the portable access point was displayed. Secure information manager **406** and/or blockchain manager **414** can then generate a credential on-demand that corresponds to the portable access point definitions and/or select a pre-generated credential that matches the portable access point definitions. In this example, secure information manager **406** has already selected the credential for the portable access point definitions in advance of receiving the credential request from requesting system **404**. When the credential request is received, the selected credential can simply be assigned to the vetted entity identified in the credential request.

In this example, a fraudulent use of this version of the user's portable access point at a later point in time can be traced back to the initial time at which this version of the user's portable access point was displayed and the vetted entity that scanned this version of the user's portable access point (and subsequently requested a credential using the deciphered information from this version of the user's portable access point). Such a fraudulent attempt may utilize a static image of the user's portable access point. A fraudster may take a photo of the user's portable access point and, using the photo, later attempt to request a credential that provides access to the user's secure information. Because

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the version of the user's portable access point identifies a time and a vetted entity that scanned the portable access point, the fraudster can be traced and held accountable.

In another example, requesting system **404** can be the user's wireless device while source information **402** can be affiliated with the vetted entity. The vetted entity can display source information **402** via any suitable display means, such as a computing system (e.g., wireless device, computer monitor, television, etc.), printed on a physical item (e.g., paper, wood, cardboard, etc.), or via any other suitable display. Source information **402** can include an access point for the vetted entity, such as a QR code or other suitable visual code affiliated with the vetted entity. The vetted entity's access point can be similar to the user's portable access point, such as the portable access point disclosed at FIG. 5.

In this example, the vetted entity's access point can include encoded information about the vetted entity, and the user's wireless device, as requesting system **404**, can scan the vetted entity's portable access point to decipher the information. The information management application at the user's wireless device, which the user interacts with to select scope permissions for the requested credential, timing for the requested credential, and/or a credential version, can be configured to decipher the vetted entity's portable access point. For example, the information management application can serve as scanning module **410**. The vetted entity's encoded information can include an identifier that identifies the vetted entity to secure information manager **406**, encoded information that represents an identity credential issued to the vetted entity (e.g., a signature using a cryptographic key), or other suitable vetted entity identifying information. Using the vetted entity information deciphered from the vetted entity's access point, the user's wireless device can generate a credential request and transmit the credential request to secure information manager **406**.

For example, the credential request generated by the user's wireless device, as requesting system **404**, can include: user identifying information (e.g., stored at the user's wireless device), scope defining information for the requested credential (e.g., selected via the information management application), timing and/or a credential version for the requested credential (e.g., selected via the information management application), vetted entity identifying information deciphered from the entity's access point, and other suitable credential request information.

In response to the credential request from the user's wireless device, secure information manager **406** can provide the vetted entity's system with access credential(s) that correspond to the credential request. For example, via interactions with blockchain manager **414**, secure information manager **406** can provide credential(s) with scope permissions that correspond to the credential request.

Blockchain manager **414** can generate a credential in response to the credential request from the user's wireless device, and this generated credential can be provided to the vetted entity's computing system. In another example, blockchain manager **414** can comprise an existing credential and blockchain manager **414** (and/or secure information manager **406**) can configure this existing credential in response to the credential request. Configuring the existing credential can include one or more of: assigning the existing credential to the vetted entity (e.g., the vetted entity's token wallet), assigning scope privileges to the existing credential that correspond to the credential request, and the like.

Using the credential(s) issued to the vetted entity's computing system via a credential request for the user's wireless

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device, as requesting system **404**, the vetted entity's computing system can issue one or more data access requests to secure information manager **406**. For example, the data access requests can include: a scope of the requested user secure information; the issued credential(s); and/or any other suitable information.

Once requesting system **404** receives the credential issued via blockchain manager **414**, requesting system **404** can issue a data request to secure information manager **406** for access to the user's secure information using the issued credential. Diagram **400B** includes requesting system **404**, secure information manager **406**, secure user information **408**, blockchain manager **414**, blockchain(s) **416**, smart contract(s) **418**, audit logs **424**, transaction logs **426**, access request **430**, and credential(s) **432**. Similar to diagram **400A**, implementations of requesting system **404** can be a system of a vetted entity, such as an entity that performed a vetting workflow by secure information manager **406**.

Requesting system **404** can issue access request **430** that comprises credential(s) **432**, which can be one or more credentials issued to requesting system **404** (and the vetted entity) by blockchain manager **414**. Access request **430** can comprise identifying information for the user, such as identifying information obtained via scanning the user's portable access point and/or the user's full name, birthdate, social security number, local address and/or zip code, medical information (e.g., primary care physician, etc.), biometric information (e.g., fingerprints, eye scan, etc.), identifier for the user's medical health records (e.g., master-patient index identifier, medical record number, insurance claim identifier, etc.), vehicle make, model, and/or license plate, representations of government issued identification (e.g., driver's license photo), images of the user's face, and the like.

Once access request **430** is received at secure information manager **406**, credential authenticator **420** can authenticate credential(s) **432** via one or more to smart contract(s) **418** of blockchain manager **418**. Secure information manager **406** can also validate the user's identifying information against registered users to confirm the request identifies a user that matches the issued credential. Once credential(s) **432** are authenticated and the user's identifying information is validated, data access controller **422** can issue one or more queries to secure user information **408** to retrieve the scope limited secure user information.

Secure information manager **408** can log the request(s), scope limited information accessed via the request(s), timestamps, and other suitable data related to access request **430** and the scope limited secure user information accessed. Secure information manager **408** can comprise an application log and/or audit log. For example, application logs can log transactions and activity of an application that permits the scope limited access in response to data access requests (e.g., access request **430**). Audit logs can log data similar to the application logs that supports auditing, for example a user audit.

The access to the user's secure information can be logged in any suitable data structure. Blockchain manager **414** can also store the audit logs in a blockchain data structure to support an audit. For example, secure information manager **408** can call one or more smart contract(s) **418** that write log information to audit logs **424**. Each request and/or instance of permitted access to the user's secure information can be logged as a block of a blockchain. Blockchain manager **414** can also store application log(s) within a blockchain data structure.

Secure information manager **408** and/or blockchain manager **414** can provide portions of the log(s) in response to an

audit request. The user (or any other suitable entity or person) can be permitted to audit the access to the user's secure information. The logged information, including the request(s), scope limited information accessed via the request(s), timestamps, and other suitable data related to the received request(s) and scope limited data can be provided to the user or other suitable audit entity.

Diagram 400C of FIG. 4C includes requesting system 404, secure information manager 406, secure user information 408, access request 430, credential(s) 432, data query 434, user information 436, scope limited user information 438, and limited user information 440. Similar to diagrams 400A and 400B, requesting system 404 can be a system of a vetted entity, such as an entity that performed a vetting workflow by secure information manager 406.

Access request 430 and credential(s) 432 can be authenticated and validated via secure information manager 406. Data query 434 can be issued to retrieve scope limited secure user information, such user information that corresponds to scope privileges of credential(s) 432. Secure user information 408 can include secure information for a number of registered persons and user information 436 can represent the secure information stored for the user who is validated against the identifying information included in access request 430 and/or the user linked to credential(s) 432. Secure information manager 406 can structure data query 434 to retrieve portions of user information 436.

User information 436 can comprise a set of data points for the user and scope limited user information 438 can comprise a subset of the data points. In an example where secure user information 408 comprises electronic health records and user health data, the set of data points of user information 438 can include an aggregate of electronic health record data for the user, and the subset of datapoints of scope limited user information 438 can comprise a portion of the aggregated electronic health data, such as metadata and/or summary data that covers components of the aggregated electronic health record data, a predefined subset of electronic health records (e.g., defined by the user, secure information manager 406, etc.), or any other suitable subset of data.

For example, scope limited user information 438 can be predefined health data points designated by the user as accessible by any suitable vetted entity that comprises authenticated credential(s). In another example, scope limited user information 438 can include health data points based on a predefined template, such as data points tailored for patient intake. Example scope limited user information 438 includes user blood type, preexisting health conditions, previous surgeries, allergies, current medications, or any other suitable user health data.

User information 436 may be a segmented health record, and the data points comprised by scope limited user information 436 can be any suitable segments of the segmented health record. For example, user information 436 can be electronic health data segmented based on parameters. Example parameters can include: originating physician and/or medical organization (e.g., entity identifier(s)), type of information (e.g., medications, tests and results, medical history, family history, biometrics, physician and patient communications, physician notes, vaccine information, allergies, etc.), relevant health practice (e.g., cardiology, primary care, neurology, oncology, etc.), date of information origination, electronic health record format, other Health Level Seven (HL7), Fast Healthcare Interoperability Resource (FHIR), and/or Substitute Medial Applications and Reusable Technologies (SMART) on FHIR data parameters,

or any other suitable health data parameters. The user can define what portions of the user's electronic health data to share via the user's portable access point by providing parameter values that define the scope. The privileges for credential(s) 432 can then be scoped based on the user defined parameters, and secure information manager 406 can limit data query 434 based on the privileges for credential(s) 432.

In response to an authenticated and validated access request 430, secure information manager 406 can issue data query 434 to secure user information 408. Secure information manager 406 can then receive limited user information 440, which can be scope limited based on the scope privileges defined for credential(s) 432. Data query 434 can be generated in response to the data requested by access request 430, and secure information manager 406 can then limit data query 434 based on the privileges of credential(s) 432. In another example, the data requested by access request 430 can be scope limited according to the privileges of credential(s) 432, and data query 434 can be generated to retrieve the scope limited data. Limited user information 440 can then be provided to requesting system 404.

role-based access control can also be applied to secure user information 408 to limit and restrict functionality and information scope relative to credential(s) 432 and the identities affiliated with the credential(s) (e.g., the user, the vetted entity, registered users of the vetted entity, etc.). Each identity affiliated with credential(s) 432 can correspond to a role in the context of the role-based access control. For example, an Administrator role may have full permissions to view all information and perform all tasks, whereas a Clerk role may only be able to view limited information about a particular consumer or patient where protected health information (PHI) or personal identifiable information (PII) attributes would be hidden from that individual. Role-based access control can also be applied to data query 434 such that limited user information 440 is retrieved by the query.

The role-based access control can be applied with respect to one or more registered identities of the vetted entity. For example, the vetted entity can be an organization and the registered identities can be persons that are part of the organization. In some embodiments, the vetted entity is a hospital, emergency room organization, or a first responder organization, and the registered identities are emergency room personnel and/or paramedics.

Request 430 may be issued by the registered identity of the vetted entity. For example, request 430 can comprise an indicator of the vetted entity and registered identity, such as credential(s) 432 that, when authenticated, identify the registered identity and vetted entity. Any other suitable indicator can be included in request 430.

Limited user information 440 can be scoped to the registered identity associated with the vetted entity that issued the request, and/or credential(s) 432 of the vetted entity or registered identity. For example, a paramedic registered identity may not provide the same level of care that an emergency room provider can provide, and therefore differing levels of information may be necessary based on the situation of care. Accordingly, a request from a paramedic registered identity may correspond to a first version of limited user information 440 while a request from emergency room personnel may correspond to a second version of limited user information 440, where the first version is different from the second version. For example, the second version can include data points relevant to certain levels and activities of care that are not included in the first version.

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Requesting system 404 can issues multiple requests, over time, to access the user's secure information using credential(s) 432. For example, as long as credential(s) 432 are unexpired and/or not revoked, secure information manager 406 can authenticate the credential(s) and permit access to the user's secure information (limited in scope to the privileges of credential(s) 432).

The user can edit the privileges assigned to credential(s) 432 via an application executing at the user's wireless device. For example, the application can be linked to secure information manager 406, and the data points, secure user information segments, and/or parameter values used group the secure user information to which credential(s) 432 are permitted access can be edited. These edits can be stored at secure information manager 406 and/or provided to the blockchain service via a software call. The blockchain service can append the privilege edits to blockchain ledger(s) that define the access privileges for credential(s) 432. The edits can include revoking the access privileges assigned to credential(s) 432. When credential(s) 432 expire or the access privileges of the credential(s) are revoked, requesting system 404 can transmit the credential(s) back to the blockchain service that issued the credential(s), for example to be assigned to the same or another vetted entity.

FIG. 5 illustrates a portable access point according to an example embodiment. Diagram 500 includes wireless device 502, portable access point 504, image 506, encoded information 508, and encoded scope sequence 510. An application running on wireless device 502 can display portable access point 504. For example, the system(s) of a vetted entity can be configured to scan portable access point 504 from wireless device 502. Wireless device 502 can display portable access point 504 in any other suitable manner.

Portable access point 504 can include image 506, or a facial image of the user, and encoded information 508. Encoded information 508 can include a unique QR code or other suitable visual or symbolic (e.g., alphanumeric) code embedded relative to image 506 (e.g., on top of image 506), such as encoded scope sequence 510. In some implementations, the QR and/or symbolic code can leverage one or more of a holographic image, a raised holographic image, pixelated code, raised texturized code, and the like. For example, encoded information 508 can be displayed on top of image 506 as an embossed QR code with a unique encrypted long string identifier that is produced using an algorithm.

Encoded information 508 can include a unique encrypted long string identifier generated via an encrypted long string identifier algorithm. The encrypted long string identifier can be unique at least due to the encrypted long string identifier algorithm that produced it. For example, if a fraudulent version of portable access point 504 attempted to replicate the unique encrypted long string identifier, the scanning of the image 506 and encoded information 508 would not match due to the unique algorithm that was used.

The unique encrypted long string identifier generated via an encrypted long string identifier algorithm can be dynamic such that a new identifier is generated for: each version of portable access point 504, over a predefined period (e.g., every 2 hours, 4 hours, 6 hours, 12 hours, daily, etc.), or via any other suitable timing. In this example, a secure information manager that manages the user's secure information can receive the dynamically generated identifier for the version of the portable access point. For example, the identifier can be dynamically generated via the information management application on wireless device 502 that man-

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ages the user's portable access points, and the application can transmit the dynamically generated identifier to the secure information manager. In another example, the identifier can be dynamically generated via a cloud service, and the cloud service can transmit the dynamically generated identifier to both the application that manages the user's portable access points and the secure information manager.

To mitigate against fraudulent attempts to request credential(s) via an old version of the user's portable access point, the secure information manager can record the one or more active long string identifier(s) for a given user. When a credential request that comprises an inactive identifier is received by the secure information manager, that request can be denied and the fraudulent attempt can be reported to support enhanced security measures.

A vetted entity system can be configured to scan portable access point 504, such as scanning module 410 and requesting system 404 of FIG. 4A. Once scanning module 410 scans portable access point 504, the requesting system 404 (e.g., an application configured to decipher portable access point 504) can decipher encoded information 508, encoded scope sequence 510, and/or image 506. For example, the application at requesting system 404 can decipher the unique long string of numbers (e.g., generated by a unique algorithm), and include the deciphered number in a credential request, as described with reference to diagram 400A of FIG. 4A.

Wireless device 502 can also display portable access point 504 while the wireless device is offline, such as while wireless device 502 lacks a connection to: the Internet, a server that hosts components of the application that displays portable access point 504, a cellular network and/or wireless local area network, or any other suitable network connection. In this example, components of the application that run locally at wireless device 502 can display portable access point 504. Portable access point 504 can be displayed on the login screen of the application, such as prior to the user logging in to the application. This can support display of portable access point 504 without a network connection. The scanning component of the vetted entity (e.g., scanning module 410) can then scan the displayed portable access point 504, and the vetted entity's system can leverage a network connection (e.g., connection to the Internet) to perform credential request(s) and/or data access request(s).

Encoded scope sequence 510 can be an encoded sequence of characters that map to portions of a user's secure information. For example, the user can define which portions of the user's secure information to share via interactions with wireless device 502 and the launched application, and the launched application can generate encoded scope sequence 510, which represent an encoded version of the user defined portions. In some implementations, a predefined mapping can map the portions of the user's secure information to the encoded scope sequence 510.

The user's secure user information linked to portable access point 504 can be electronic health data segmented based on parameters. For example, the parameters can include: originating or attributed physician and/or medical organization (e.g., entity names or identifier(s)), type of information (e.g., medications, tests and results, medical history, family history, biometrics, physician and patient communications, physician notes, vaccine information, allergies, etc.), relevant health practice (e.g., cardiology, primary care, neurology, oncology, etc.), date of information origination, electronic health record format, other Health Level Seven (HL7), Fast Healthcare Interoperability Resource (FHIR), and/or Substitute Medial Applications and

Reusable Technologies (SMART) on FH IR data parameters, or any other suitable health data parameters. The user can define what portions of the user's electronic health data to share via portable access point **504** by providing parameter values that define the scope.

For example, via interactions with the application executing at the user's wireless device, the user can specify the following parameter values: originating physician and/or medical organization—ALL; types—medications, tests and results, medical history, family history, biometrics, vaccine information, and allergies; relevant health practices—cardiology, primary care physician, and neurology; date of information origination—ALL; and electronic health record format—ALL. Other example parameter values for date of information include past two years, past year, since the age of 18, custom time range (e.g., Jan. 1-31, 2023), etc. The user's electronic health data that matches the parameter values specified by the user can be scoped for sharing via portable access point **504** and any credential(s) issued that correspond to portable access point **504**.

A predefined mapping can map the user's health data segments to the encoded scope sequence **510**. In an example, a formatting for the sequence can define which sequence of characters map to a parameter and which sequence of characters map to values for that parameters. A sample encoded scope sequence **510** comprises [A:XYX, C:1C3B, X:1456, etc.]. In an example predefined mapping, the initial symbols of the encoded scope sequence can map to one of the electronic health data segments (e.g., originating or attributed physician and/or medical organization, type of information, relevant health practice, date of information origination, electronic health record format, etc.). In this example, predefined mapping, the symbols after the ':' value can map to parameters values.

In an example where the letter 'A' maps to date of information origination in the predefined mapping, the symbols 'XYX' can map to the parameters value 'Information from the last 5 years'. Other symbols can map to other date of information origination parameters values, such as 'ALL', 'since the age of 18', or custom date ranges. In an example where the letter 'C' maps to types of information in the predefined mapping, the symbols '1C3B' can map to a subset of information types (e.g., medications, tests and results, medical history, family history, biometrics, vaccine information, and allergies). Other symbols can map to other subsets of types of information.

In an example where the letter 'X' maps to relevant health practices in the predefined mapping, the symbols '1456' can map to a subset of health practices (e.g., cardiology, primary care, neurology, and oncology). Other symbols can map to other subsets of health practices. In this example, encoded scope sequence **510** can define parameters of a user's electronic health data and values for those parameters. When portable access point **504** is scanned by a vetted entity, such as scanning module **410** and requesting system **404** of FIG. **4A**, requesting system **404** (e.g., an application configured to decipher portable access point **504**) can decipher encoded scope sequence **510**. For example, the application at requesting system **404** can decipher encoded scope sequence **510**, and include the deciphered sequence of symbols (e.g., [A:XYX, C:1C3B, X:1456, etc.]) in a credential request, as described with reference to diagram **400A** of FIG. **4A**.

A secure information manager can issue a credential to requesting system **404** (via a blockchain service) with privileges that correspond to the sequence of symbols. For example, secure information manager **406** and blockchain manager **414** of FIG. **4A** can communicate to issue a

blockchain backed credential comprising privileges defined by the sequence of symbols. Secure information manager **406** can store the privileges and/or blockchain manager **414** can store the privileges relative to the issued credential as part of the blockchain(s) **416**.

Secure information manager **406** can also receive data access requests comprising an issued credential from requesting system **404**, where the data access request comprises requested segments of the user's electronic health records. For example, requesting system **404** can issue access request **430** comprising credential(s) **432**, as illustrated in FIG. **4B**. Access request **430** can also comprise a scope of requested data, such as defined portions (e.g., segments and parameter values) of the user's electronic health records. The scope of requested data can be defined using a sequence of symbols that corresponds to a predefined mapping (similar to encoded scope sequence **510**). For example, secure information manager **406** can interpret the sequence of symbols using the predefined mapping to determine the requested portions of the user's electronic health data (e.g., segments limited by parameter values).

Secure information manager **406** can compare the access privileges of the credential(s) **432** (e.g., sequence of symbols interpreted via a predefined mapping) and the scope of requested data (e.g., another sequence of symbols interpreted via the same or another predefined mapping) to determine which segments and which portions of each segment (as defined by parameter values) of the user's electronic health data to retrieve for access request **430**. For example, the access privileges of the credential(s) **432** may be permitted to access only a subset of the scope of requested data.

Audit logs **424** of blockchain manager **414** can publish, at a private blockchain of blockchain(s) **416**, one or more predefined mappings that map sequence(s) of symbol to segments of a user's electronic health records and parameters values for those segments. For example, secure information manager **406** can access the published predefined mappings to interpret credential privileges and/or data access requests. In another example, an application loaded to a user's wireless device that manages the user's portable access point(s) can access the published predefined mappings to encode the user's portable access point.

The predefined mappings can be updated, for example based on formatting changes for electronic health records, changes to a user's electronic health record, or for any other suitable cause. Blockchain manager **414** can append a new block to the private blockchain that stores the predefined mapping(s) such that: the updated mapping is stored as the latest block, and the historic mappings are stored as previous blocks. In this example, participating computer systems can access the current predefined mappings via the latest block.

In addition, audit logs **424** can store data access requests and one or more sequences of symbols for the data access requests, such as the scope of the requested data, the privileges of the credential used to access the user's electronic health records, and the actual electronic health record segments accessed via a data access request.

FIG. **6** illustrates a flow diagram for permitting limited access to a user's secure information using credential authentication and user verification according to an example embodiment. In one embodiment, the functionality of FIG. **6** (and FIGS. **7** and **8** below) is implemented by software stored in memory or other computer-readable or tangible medium, and executed by a processor. In other embodiments, each functionality may be performed by hardware (e.g., through the use of an application specific integrated

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circuit (“ASIC”), a programmable gate array (“PGA”), a field programmable gate array (“FPGA”), etc.), or any combination of hardware and software.

Process 600 can be performed by a requesting computing system, such as a computing system affiliated with a vetted entity that issues requests (e.g., credential requests and/or data access requests) to a secure information manager. Process 602 can be performed by a secure information manager (e.g., cloud computing system or any other suitable computing system) that manages secure information for user(s).

At block 604, process 600 can scan a portable access point. For example, a requesting system can include a scanning component configured to scan and decipher a user’s portable access point. The scanning component can be a portable device that comprises a camera and software configured to decipher user identifying information from the user’s portable access point.

A portable access point can be visual access point that, when scanned, can permit access to a user’s secure information (e.g., stored and managed via a secure information manager). For example, the visual access point can comprise a visual representation of the user linked to the portable access point (e.g., facial image) and encoded information related to the scope of the user’s secure information and/or permitted access. The requesting system and scanning component can be affiliated with a vetted entity, such as an entity that performed a vetting workflow. Example vetted entities can include service providers (e.g., system administrators, health care providers, hospitals, etc.), partners in a joint venture, a person registered with the secure information manager, or any other suitable vetted entity.

The user can configure the portable access point and the encoded information displayed. For example, the portable access point can be displayed via an application executing at the user’s wireless device (e.g., smartphone, tablet, etc.) and the user can interact with the application and select a sharing scope for the user’s secure information. The portable access point can be dynamic such that the user’s selections via the application generate different versions of the portable access point with different encoded information displays. For example, the user can define a sharing scope that identifies data points of the user’s secure information that can be shared with a vetted entity via scanning of the portable access point. The user can also define a sharing scope for a time period over which the user’s secure information can be shared with a vetted entity via scanning of the portable access point. Process 700 of FIG. 7 further describes user interactions to define timing parameters and scope parameters for configuring a portable access point.

At block 606, process 600 can generate a credential request in response to scanning the portable access point. For example, the requesting system can decipher the user’s portable access point and generate a request for one or more credentials using the deciphered information. The encoded information displayed by the portable access point can be deciphered and included in the credential request. Deciphered information from the user’s portable access point can include user identifying information, such as full name, birthdate, social security number, local address and/or zip code, medical information (e.g., primary care physician, etc.), biometric information (e.g., fingerprints, eye scan, etc.), identifier for the user’s medical health records (e.g., master-patient index identifier), vehicle make, model, and/or license plate, representations of government issued identification (e.g., driver’s license photo), images of the user’s face, and the like.

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Deciphered information from the user’s portable access point can also include a scope definition corresponding to the user’s secure information. For example, the scope definition can correspond to a credential type, such as a vanishing credential (limited scope of time), episodic credential (predefined scope of time), or a durable credential (valid until revoked). In another example, the scope definition can correspond to limited scope privileges for the requested credential.

The user’s secure information can be electronic health data segmented based on parameters, and the limited scope privileges for the requested credential can correspond to limited portions of the user’s electronic health data. For example, the parameters can include: originating physician and/or medical organization (e.g., entity identifier(s)), type of information (e.g., medications, tests and results, medical history, family history, biometrics, physician and patient communications, physician notes, vaccine information, allergies, etc.), relevant health practice (e.g., cardiology, primary care, neurology, oncology, etc.), date of information origination, electronic health record format, other Health Level Seven (HL7), Fast Healthcare Interoperability Resource (FHIR), and/or Substitute Medial Applications and Reusable Technologies (SMART) on FHIR data parameters, or any other suitable health data parameters. The user can define what portions of the user’s electronic health data to share via the user’s portable access point by providing parameter values that define the scope.

At block 608, process 600 can transmit the credential request. For example, the requesting system can transmit the credential request to a secure information manager. The requesting system’s credential request can be for credential(s) that permit the requesting system access to the user’s secure information, which can be managed by the secure information manager. For example, the details of the request (e.g., user identifying information, credential type, defined scope secure user information privileges, etc.) can be provided by the user’s portable access point, which permits the user to control how the user’s secure information is shared with the requesting system.

The secure information manager can validate and authenticate the credential request and obtain a credential from a blockchain service in response to the request. The authentication and validation of the credential request and subsequent retrieval of the credential from the blockchain service are described with reference to blocks 618 and 620 of process 602. A block 610, process 600 can receive credential(s). For example, the secure information manager can transmit the obtained credential to the requesting system.

At block 612, process 600 can transmit a data access request that comprises the received credential(s). For example, a data access request that includes the received credentials, an identifier for the vetted entity that issued the request, and user identifying information can be transmitted to the secure information manager. The secure information manager can validate and authenticate the data access request and retrieve scope limited secure user information in response to the data access request.

The data access request can define one or more data points of the user’s secure information. For example, the user’s secure information can be electronic health data segmented based on parameters. The data access request can include specific parameter values that define the scope of the user’s secure information requested. In some implementations, the credential issued to the vetted entity and provided in the data access request comprises scope privileges relative to the user’s secure information. The authentication and validation

of the data access request and subsequent retrieval of the scoped limited secure user information are described with reference to blocks **626** and **628** of process **602**.

At block **614**, process **600** can receive scope limited secure user information in response to the data access request. For example, the secure information manager can transmit the scope limited user information to the requesting system. The received scope limited user information can correspond to the requested data point(s) included in the data access request. The received scope limited user information can also correspond to a portion of the requested data point(s) included in the data access request. For example, where the data access request includes user data point(s) outside the set of scope privileges of the vetted entity/provided credential(s), the secure information manager may retrieve and return only the portion of the user data point(s) covered by the set of scope privileges.

At block **616**, process **602** can receive the credential request. For example, the secure information manager can receive the credential request from the requesting system, as described with reference to block **608** of process **600**. The credential request can include user identifying information, an identifier and/or credential of a vetted entity, credential type definition, scope definitions for secure user information, and any other suitable information.

The secure information manager can validate the user identifying information and authenticate the vetted entity identifier/credential. In this example, secure information manager can obtain a credential that permits the vetted entity scope limited access to the user's secure information (e.g., from a blockchain service). The obtained credential can correspond to the credential type definition from the credential request and the access privileges assigned to the obtained credential can correspond to the scope definitions from the credential request.

At block **618** process **602** can authenticate and validate the credential request. For example, the credential request can include user identifying information that the secure information manager validates. The secure information manager can manage secure information for a number of registered persons. The validation can match the user identifying information provided in the credential request to one of the registered persons.

The credential request can include an identifier and/or identifying credential of a vetted entity and the secure information manager can authenticate the identifier and/or identifying credential. For example, identifier and/or identifying credential included in the credential request (from a requesting system affiliated with the vetted entity) can include a unique identifier assigned to the vetted entity, cryptographic key and/or digital signature generated via a cryptographic key assigned to the vetted entity, a unique token assigned to the vetted entity, or any other suitable data that can be used to authenticate the vetted entity.

At block **620**, process **602** can obtain credential(s) corresponding to the credential request. The validation of the user identifying information can match a registered user to the credential request and the authentication of the identifier/credential can match a vetted entity to the credential request. The secure information manager can then issue a software call to a blockchain service for a credential with scope privileges that permit the matching vetted entity scope limited access to the matching registered user's secure information. For example, the software call can include an identifier for the registered user, an identifier for the vetted entity, and the credential type included in the credential request from the requesting system.

The blockchain service can return a blockchain backed credential to the secure information manager in response to the software call. For example, the blockchain service can host one or more blockchains that manage credentials, and the credentials can be issued to vetted entities to serve as credentials that permit the vetted entities access to secure user information. The issued credential can match the credential type included in the software call. The transaction issuing the credential to the vetted entity can be appended to the blockchain(s) such that a record of assignment is maintained as an immutable ledger.

The software call by the secure manager can be a smart contract call and the blockchain service can execute one or more smart contracts to return the credential in response to the smart contract call. For example, the smart contract execution can issue the credential to a token well identifier associated with the vetted entity (or any other suitable entity identifier) and append the transaction to the managed blockchain(s). The appended transaction can include additional information, such as an expiration timer for the issuance of the token to the vetted entity (e.g., defined by the credential type), identifying information about the vetted entity, identifying information about the user, or any other suitable information.

The secure information manager can assign the scope sharing privileges (e.g., scope of secure user information data point(s)) included in the credential request to the credential. For example, the data points, secure information segments, or other suitable portions of the user's secure information defined via the scope sharing definitions can be assigned as privileges of the credential returned by the blockchain service and the privilege assignment(s) can be stored by the secure information manager.

The software call from the secure information manager to the blockchain service for the credential can include the scope sharing definitions, and the blockchain service can assign the credential the corresponding privileges to access the user's secure information. For example, the blockchain service can append the scope sharing definitions (via execution of smart contract(s)) to the blockchain(s) as part of the recorded transaction(s).

At block **622**, process **602** can transmit the credential(s). For example, the secure information manager can send the obtained credential(s) to the requesting system. The blockchain service can also transmit the credential(s) directly to the requesting system via a separate communication channel.

At block **624**, process **602** can receive a data access request. For example, the secure information manager can receive a data access request with one or more credentials from the requesting system, as described with reference to block **612** of process **600**. The data access request can include a credential, an identifier for the vetted entity that issued the request, user identifying information, one or more requested data points of a user's secure information, and any other suitable information.

At block **626**, process **602** can authenticate and validate the data access request. For example, the data access request can include a credential. The secure information manager can authenticate the credential via a software call to the blockchain service. For example, the blockchain service can return trusted information about the credential from the blockchain(s) that record transactions for the credential, such as: the vetted entity assigned the credential, the registered user to which the credential privileges apply, the expiration time for the credential, the data points that the credential is privileged to access, or any other transaction

information stored by the blockchain(s). The secure information manager can authenticate the data access request when the expiration timer has not expired and the vetted entity assigned the credential matches the identifier for the vetted entity that issued the request.

The data access request can include an identifier for the vetted entity that issued the request, and the secure information manager can validate that the identifier for the vetted entity included in the data access request matches the vetted entity returned by the blockchain service. The data access request can also include user identifying information that identifies a user, and the secure information manager can validate that the user information matches: one of the registered persons managed by the secure information manager, the user affiliated with the credential returned by the blockchain service, or any combination thereof.

At block 628, process 602 can retrieve scope limited secure information in response to the data access request. For example, the secure information manager can query a secure data store that stores the user's secure information for the user information requested in the data access request.

In response to an authenticated and validated data access request, the secure information manager can generate a data query that retrieves scope limited secure user information from a secure data store. For example, the data query can be scope limited based on the scope privileges assigned to the authenticated credential. In some implementations, the data query can be generated in response to the data requested by access request, and the secure information manager can then limit the data query based on the privileges of the provided credential. In another example, the secure user information requested by the data access request can be scope limited according to the privileges of the provided credential, and the data query can be generated to retrieve the scope limited data. The secure data store can then return the scope limited secure user information in response to the data query.

At block 630, process 602 can provide the scope limited secure user information to the requesting system. For example, the secure information manager can return the secure user information requested by the data access request or a further scope limited version of the secure user information requested by the data access request, such as when the provided credential does not have access privileges for the entirety of the secure user information requested by the data access request.

FIG. 7 illustrates a flow diagram for defining access limitations for sharing a user's secure information according to an example embodiment. Process 700 can be performed by a user system, such as a wireless device (e.g., smartphone, tablet, etc.). Process 702 can be performed by a requesting system, such as a system of a vetted entity that issues a credential request to a secure information manager. Process 700 can be triggered when a user launches an information management application at the user system that manages scope definitions for sharing the user's secure information. Process 702 can be triggered when the requesting system scans a display of the user system, such as a portable access point displayed via the user system.

At block 704, process 700 can receive timing parameters for sharing the user's secure information. For example, the user can provide the timing parameters via the user system and the application launched at the user system. The user can define timing parameters by selecting a type of credential, such as vanishing, episodic, and/or durable, each of which corresponds to a type of timing parameter. The user can define any other suitable timing parameters.

At block 706, process 700 can receive scope parameters for sharing the user's secure information. For example, the user can provide the scope parameters via the user system and the information management application launched at the user system. The scope parameters can define portions of the user's secure information. The user's secure information may comprise electronic health records, and the scope parameters can define parameters of the user's electronic health records and parameter values.

For example, the parameters that segment the user's electronic health records can include: originating physician and/or medical organization (e.g., entity identifier(s)), type of information (e.g., medications, tests and results, medical history, family history, biometrics, physician and patient communications, physician notes, vaccine information, allergies, etc.), relevant health practice (e.g., cardiology, primary care, neurology, oncology, etc.), date of information origination, electronic health record format, other Health Level Seven (HL7), Fast Healthcare Interoperability Resource (FHIR), and/or Substitute Medial Applications and Reusable Technologies (SMART) on FHIR data parameters, or any other suitable health data parameters. One or more of these electronic health record parameters can be further scoped according to parameters values (e.g., dates, subsets of information types, subsets of relevant health practices, etc.). The user can define what portions of the user's electronic health data to share via the information management application by providing parameter values that define the scope.

At block 708, process 700 can generate encoded information based on the timing parameters and scope parameters. For example, the encoded information can encode a credential type (representing the timing for the credential) and scope parameters (representing portions of the user's secure information). A predefined mapping can map the user's health data segments (e.g., parameters and parameter values) to encoded sequences of characters/symbols. In an example, a formatting for the sequence can define which sequence of characters map to a parameter and which sequence of characters map to values for that parameters.

At block 710, process 700 can display a portable access point that includes the encoded information. For example, the displayed portable access point can include an image of the user, user identifying information (some or all of which can be encoded), and the generated encoded information that represents the credential type and the scope parameters.

At block 712, process 702 can scan the portable access point. For example, a scanning module of the requesting system (e.g., computing system with a camera and application configured to decipher the portable access point) can scan the portable access point displayed via the user system. At block 714, process 702 can decipher the encoded information from the portable access point. For example, the scanning module of the requesting system can be configured to decipher the portable access point, such as extracting information displayed by the portable access point, and/or deciphering encoded information displayed by the portable access point. The deciphered information can include the timing parameters and the scope parameters defined by the user via the information management application.

At block 716, process 702 can generate a credential request using the extracted and/or deciphered information. For example, the credential request can request a credential affiliated with one or more pieces of information extracted and/or deciphered from scanning the portable access point. The credential request can include user identifying information, a credential type and/or timing parameters, scope

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parameters (e.g., sequence of characters/symbols that map to portion(s) of the user's electronic health records), and other suitable information (e.g., entity identifying information).

At block 718, process 702 can transmit the credential request to a secure information manager that manages the user's secure information. For example, the secure information manager can authenticate and/or validate the credential request and issue a credential to the requesting system (via interactions with a blockchain service) in response to the credential request.

FIG. 8 illustrates a flow diagram for retrieving scope limited user information from a secure data store and logging the access according to an example embodiment. Process 800 can be performed by a secure information manager that has permitted a vetted entity access to a user's secure information, for example in response to a request from the vetted entity's computing system. Process 800 can be performed by one or more components of a secure data store. In another example, process 800 can be performed by one or more components of blockchain service. In various embodiments, process 800 can be performed at the secure information manager, secure data store, blockchain service, or any combination thereof.

At block 802, process 800 can query a secure data store for the user's secure information. For example, a vetted entity can request a set of data points for a user's secure information. A data store query (e.g., SQL, any other suitable database query) can be generated that queries for all or a portion of the vetted entity's requested data points of the user's secure information. For example, the data store query can be limited to the privileges of the vetted entity (e.g., a credential assigned to the vetted entity).

At block 804, process 800 can receive scope limited user information from the secure data store. For example, the scope limited user information can be limited by the formulated data store query or by the data store itself. The generated query is limited to request a limited scope of the user's secure information that the vetted entity is permitted to access. The data store can return scope limited data points of the user's secure information that are limited to the data points that the vetted entity is permitted to access.

At block 806, process 800 can provide the scope limited user information to the requesting system of the vetted entity. For example, the returned scope limited data points of the user's secure information can be transmitted to the vetted entity's computing system that issued the request.

At block 808, process 800 can log the request(s), credential(s), scope limited information accessed via the request(s), timestamps, and other suitable data related to the received request(s) and scope limited data. For example, the access to the user's secure information can be logged in any suitable data structure. The storage structure can be a blockchain, and each request and/or instance of permitted access to the user's secure information is logged as a block of the blockchain. Example log information can include: the scope of the data access requested (e.g., parameters and parameters values of the user's secure information requested, sequence of characters/symbols that represent such parameters and parameter values, etc.), the provided credential(s) and permissions (e.g., timing parameters, scope parameters, such as secure information parameters and parameters values and/or the sequence of characters/symbols that represent such parameters and parameter values, etc.), timestamps, entity identifier(s) (and in some examples person identifier(s) affiliated with the entity), user identifying information, secure user information provided in response to the request(s) (e.g.,

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parameters and parameters values that define the portions of accessed data), and any other suitable information.

One or more predefined mappings can be affiliated with the data access request logs. For example, the predefined mappings can map sequences of characters/symbols to portions of the user's secure information (e.g., parameters and parameters values of electronic health data). The predefined mappings can be published to a private blockchain such that the mappings can be used to interpret credential privileges and/or data access requests that comprise sequences of characters/symbols affiliated with the mappings.

The predefined mappings can be updated, for example based on changes to the user's secure information (e.g., formatting changes to the user's electronic health records, new user electronic health records, etc.). In response to an update, a new block can be appended to the private blockchain that stores the predefined mapping(s) such that: the updated mapping is stored as the latest block, and the historic mappings are stored as previous blocks. In this example, participating computer systems can access the current predefined mappings via the latest block.

At block 810, process 800 can provide the logged data in response to an audit request. The user (or any other suitable entity or person) can be permitted to audit the access to the user's secure information. The logged information, including the request(s), scope limited information accessed via the request(s), timestamps, and other suitable data related to the received request(s) and scope limited data can be provided to the user or other suitable audit entity.

In an example where the logged data for an access request includes sequence(s) of characters/symbols, the private blockchain that manages the predefined mappings can be accessed to interpret the sequence(s) of characters/symbols. Once interpreted, the portions of the user's secure information (e.g., parameter and parameters values of the user's electronic health data) represented by the sequence(s) of characters/symbols can be provided in response to the audit request. For example, the timestamp for a given access request can be used to determine the given predefined mapping that was active during the given access request. This determined predefined mapping can be accessed via the private blockchain to interpret any sequence(s) of characters/symbols logged for the given access request.

The portions of the user's secure information represented by the sequence(s) of characters/symbols (with respect to the determined predefined mapping) can be provided in response to an audit request that covers the given access request. For example, the logged data provided for the given access request in response to the audit can include: the scope of the data access requested (e.g., parameters and parameters values of the user's secure information requested), the provided credential(s) and permissions (e.g., timing parameters, scope parameters, such as secure information parameters and parameters values, etc.), timestamps, entity identifier(s) (and in some examples person identifier(s) affiliated with the entity), user identifying information, secure user information provided in response to the request(s) (e.g., parameters and parameters values that define the portions of accessed data), and any other suitable information.

A user audit of the logged access to the user's secure user information via the secure information manager can be achieved using an information management application (e.g., smartphone application, web application, etc.). For example, the information management application can sort and display the logged access by: vetted entity; date and time of access; and the like. The information management application can list "instances" of access as, such as by date and

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time, and the user can select the instance of access to expand the specific details of the access (e.g., vetted entity/registered identity, scope of access, etc.). In some embodiments, the information management application can log instances/versions of the user's portable access point and associate a given instance/version of a portable access point with each logged access by a vetted entity. This can aid the user's inspection for unauthorized access attempts.

A privacy enforcement entity can also audit logs with respect to individual users, individual vetted entities, groups of users, and/or groups of vetted entities. For example, the secure information manager (or some other privacy enforcement entity) can periodically review the logged access to users' secure user information by vetted entities. When a user reports unauthorized access by a given vetted entity, the privacy enforcement entity can audit the vetted entity's access across multiple users to uncover other instances of unauthorized access by the vetted entity. For example, the secure information manager can revoke the "vetted" status of the vetted entity when a pattern of unauthorized access is detected.

Embodiments permit scope limited access to a user's secure information using blockchain backed credential(s). A user can register with a secure information manager and control the scope with which the user's secure information is shared. For example, the user can permit a vetted entity (e.g., service provider, health care provider, other individual, etc.) access to the user's secure information via a portable access point. The user can select scope definition that control how the user's secure information is shared with the vetted entity. The vetted entity can scan the user's portable access point and request a credential that permits access to the user's secure information via the scanning. For example, the credential can be a blockchain backed credential that is assigned access privileges that correspond the user's selections.

The vetted entity can then issue one or more data access requests using the credential. For example, the data access request(s) can be authenticated and validated by the secure information manager. The secure information manager can permit the vetted entity scope limited access to the user's secure information (based on an authenticated and validated data access request) that corresponds to the access privileges assigned to the credential. The user can revoke the access privileges assigned to the credential and/or vetted entity at any time. The access privileges assigned to the credential can include an expiration timer, after which the credential will no longer authenticate with the secure information manager.

Embodiments achieve fine grained user controlled access to the user's secure information that is efficient and secure. For example, the user's portable access point is configured to efficiently define sharing conditions for the user's secure information. In addition, the issued credentials and secure information manager enforce the user's sharing conditions in a trusted manner. In embodiments where the credential is blockchain backed, the blockchain based management ensures that the credential is authentic and mitigates against fraudulent attempts to access the user's secure information.

The features, structures, or characteristics of the disclosure described throughout this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of "one embodiment," "some embodiments," "certain embodiment," "certain embodiments," or other similar language, throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the embodiment

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may be included in at least one embodiment of the present disclosure. Thus, appearances of the phrases "one embodiment," "some embodiments," "a certain embodiment," "certain embodiments," or other similar language, throughout this specification do not necessarily all refer to the same group of embodiments, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

One having ordinary skill in the art will readily understand that the embodiments as discussed above may be practiced with steps in a different order, and/or with elements in configurations that are different than those which are disclosed. Therefore, although this disclosure considers the outlined embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent, while remaining within the spirit and scope of this disclosure. In order to determine the metes and bounds of the disclosure, therefore, reference should be made to the appended claims.

What is claimed is:

1. A method for permitting limited access to a user's secure information, the method comprising:

receiving, at a secure information manager from a requesting system, a credential request for one or more credentials that permit access to a user's secure information, the credential request comprising user identifying information, entity identifying information, and scope definitions, wherein the requesting system generates the credential request in response to scanning a portable access point of a user;

validating, at the secure information manager, the user identifying information and the entity identifying information;

assigning, to the entity and in response to the validating, a blockchain credential with access privileges that correspond to the scope definitions, wherein the blockchain credential assignment is recorded on a private blockchain that manages the blockchain credential; and permitting, in response to one or more access requests from the entity that comprises the assigned blockchain credential, access to the user's secure information limited to the access privileges of the blockchain credential.

2. The method of claim 1, wherein, the portable access point for the user comprises encoded information, in response to scanning the portable access point, the requesting system is configured to decipher the user identifying information and scope definitions from the encoded information.

3. The method of claim 2, wherein, the user provides selections via input at a portable device, the portable device is configured to generate the portable access point in response to the user selections, and the requesting system scans the portable access point from the portable device.

4. The method of claim 3, wherein, the user selections correspond to segment identifiers that define segments of the user's secure information selected by the user for sharing with the entity, the encoded information of the portable access point comprises encoded representations of the segment identifiers,

the requesting system is configured to decipher the encoded representations of the segment identifiers in response to scanning the portable access point, and

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the scope definitions provided in the credential request comprise the deciphered segment identifiers.

5. The method of claim 4, further comprising:

mapping the scope definitions that comprise the deciphered segment identifiers to segments of the user's secure information using a predefined mapping, wherein the access privileges of the assigned blockchain credential permit the entity access to the mapped segments of the user's secure information.

6. The method of claim 5, further comprising:

storing, using one or more blockchains of a blockchain manager, one or more logs of the scope limited access to the user's secure information, the logs comprising one or more of: an identifier for the entity, portions of the access requests, segment identifiers for the user's secure information accessed by the entity, timestamps for the accessing of the user's secure information, or any combination thereof, wherein the one or more logs are recorded as blocks of the one or more blockchains.

7. The method of claim 6, further comprising:

storing, using the one or more blockchains of the blockchain manager, the predefined mapping, wherein the predefined mapping comprises an active mapping from among a plurality of predefined mappings recorded as blocks of the one or more blockchains.

8. The method of claim 7, further comprising:

providing, in response to an audit request, audit information descriptive of scope limited access to the user's secure information over a time period, wherein the audit information is generated by: accessing the logs of the scope limited access to the user's secure information over the time period and the stored plurality of predefined mappings, and translating mapped segment identifiers from the accessed logs using the stored predefined mappings.

9. The method of claim 1, wherein the entity comprises a vetted entity that is vetted after performance of a vetting workflow.

10. The method of claim 1, wherein permitting scope limited access to the user's secure information further comprises:

receiving, at a secure information manager, the one or more access requests, the access requests comprising an unauthenticated credential;

authenticating, via the private blockchain, the unauthenticated credential as the blockchain credential assigned to the entity; and

permitting, in response to the authenticating, the access to the user's secure information limited to the access privileges of the blockchain credential.

11. A non-transitory computer readable medium having instructions stored thereon that, when executed by a processor, cause the processor to permit limited access to a user's secure information, wherein, when executed, the instructions cause the processor to:

receive, at a secure information manager from a requesting system, a credential request for one or more credentials that permit access to a user's secure information, the credential request comprising user identifying information, entity identifying information, and scope definitions, wherein the requesting system generates the credential request in response to scanning a portable access point of a user;

validate, at the secure information manager, the user identifying information and the entity identifying information;

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assign, to the entity and in response to the validating, a blockchain credential with access privileges that correspond to the scope definitions, wherein the blockchain credential assignment is recorded on a private blockchain that manages the blockchain credential; and permit, in response to one or more access requests from the entity that comprises the assigned blockchain credential, access to the user's secure information limited to the access privileges of the blockchain credential.

12. The non-transitory computer readable medium of claim 11, wherein,

the portable access point for the user comprises encoded information,

in response to scanning the portable access point, the requesting system is configured to decipher the user identifying information and scope definitions from the encoded information.

13. The non-transitory computer readable medium of claim 12, wherein,

the user provides selections via input at a portable device, the portable device is configured to generate the portable access point in response to the user selections, and the requesting system scans the portable access point from the portable device.

14. The non-transitory computer readable medium of claim 13, wherein,

the user selections correspond to segment identifiers that define segments of the user's secure information selected by the user for sharing with the entity,

the encoded information of the portable access point comprises encoded representations of the segment identifiers,

the requesting system is configured to decipher the encoded representations of the segment identifiers in response to scanning the portable access point, and the scope definitions provided in the credential request comprise the deciphered segment identifiers.

15. The non-transitory computer readable medium of claim 14, wherein the instructions further cause the processor to:

map the scope definitions that comprise the deciphered segment identifiers to segments of the user's secure information using a predefined mapping, wherein the access privileges of the assigned blockchain credential permit the entity access to the mapped segments of the user's secure information.

16. The non-transitory computer readable medium of claim 15, wherein the instructions further cause the processor to:

store, using one or more blockchains of a blockchain manager, one or more logs of the scope limited access to the user's secure information, the logs comprising one or more of: an identifier for the entity, portions of the access requests, segment identifiers for the user's secure information accessed by the entity, timestamps for the accessing of the user's secure information, or any combination thereof, wherein the one or more logs are recorded as blocks of the one or more blockchains.

17. The non-transitory computer readable medium of claim 16, wherein the instructions further cause the processor to:

store, using the one or more blockchains of the blockchain manager, the predefined mapping, wherein the predefined mapping comprises an active mapping from among a plurality of predefined mappings recorded as blocks of the one or more blockchains.

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18. The non-transitory computer readable medium of claim 17, wherein the instructions further cause the processor to:

provide, in response to an audit request, audit information descriptive of scope limited access to the user's secure information over a time period, wherein the audit information is generated by: accessing the logs of the scope limited access to the user's secure information over the time period and the stored plurality of predefined mappings, and translating mapped segment identifiers from the accessed logs using the stored predefined mappings.

19. The non-transitory computer readable medium of claim 11, wherein the entity comprises a vetted entity that is vetted after performance of a vetting workflow.

20. A system for permitting limited access to a user's secure information, the system comprising:

a processor; and

a memory storing instructions for execution by the processor, the instructions configuring the processor to: receive, at a secure information manager from a requesting system, a credential request for one or

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more credentials that permit access to a user's secure information, the credential request comprising user identifying information, entity identifying information, and scope definitions, wherein the requesting system generates the credential request in response to scanning a portable access point of a user;

validate, at the secure information manager, the user identifying information and the entity identifying information;

assign, to the entity and in response to the validating, a blockchain credential with access privileges that correspond to the scope definitions, wherein the blockchain credential assignment is recorded on a private blockchain that manages the blockchain credential; and

permit, in response to one or more access requests from the entity that comprises the assigned blockchain credential, access to the user's secure information limited to the access privileges of the blockchain credential.

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